

LiquidSR: Lightweight image super-resolution via liquid ODE-based adaptive feature update

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ARTICLE INFO

Keywords:

Lightweight image super-resolution

Liquid neural networks

ODE

Adaptive feature update

ABSTRACT

Most existing lightweight image super-resolution (SR) methods rely on a static feature mapping paradigm, which cannot adaptively accommodate the spatial heterogeneity of texture complexity. Under tight parameter and computational budgets, this inherent static nature imposes a fundamental trade-off between restoration fidelity and inference efficiency. In this work, we propose LiquidSR, a lightweight SR framework that introduces an Ordinary Differential Equation (ODE)-driven adaptive feature update mechanism inspired by Liquid neural networks (LNNs) to overcome the limitations of static modeling. Specifically, we design a Spatial-aware Spatial Heterogeneity Neuron (SSN) module that models feature evolution via ODEs, where learnable per-channel time constants τ assign distinct update rates to different feature channels. To handle the multi-time-step features generated by this continuous evolution, we propose an Adaptive Temporal Attention Weights (ATAW) module that enhances informative time steps and suppresses redundancy. Furthermore, we introduce a Differentiable Spatial Heterogeneity Self-Attention (DSSA) module that optimizes spatial heterogeneity representation via ODEs prior to self-attention, thereby synergizing continuous feature update with long-range dependency modeling. Extensive experiments on five standard benchmarks demonstrate that LiquidSR achieves competitive state-of-the-art performance among lightweight SR methods, with a well-balanced trade-off between reconstruction quality and inference efficiency. The code is available at <https://github.com/HZAI-ZJNU/LiquidSR>.

1. Introduction

Image Super-Resolution (SR) (Chen et al., 2023; Huang et al., 2015; Ledig et al., 2017; Lim et al., 2017; Qiu et al., 2019; Rad et al., 2019; Song et al., 2021; Timofte et al., 2016) is a classic low-level computer vision task that reconstructs high-resolution (HR) images from their low-resolution (LR) counterparts by recovering lost high-frequency details. It has broad applications in surveillance systems (Gómez-Bautista & Calderón-Bocanegra, 2024), medical imaging (Lu et al., 2025; Sarkar et al., 2022; Zhao et al., 2026), and remote sensing (Hsu & Chen, 2022; Tang et al., 2026; Wang et al., 2025; Xie et al., 2024), with consid-

erable practical value. With the growing demand for deployment on resource-constrained platforms such as mobile devices and edge nodes, lightweight SR (Ahn et al., 2018; Hui et al., 2019, 2018; Li et al., 2020; Liu et al., 2020; Luo et al., 2020; Zamir et al., 2022) has attracted increasing research attention. The core challenge in this field is to preserve reconstruction quality under strict parameter and computational constraints. However, existing lightweight methods are limited by their static modeling paradigm, which restricts the achievable balance between performance and efficiency.

Image texture restoration is inherently characterized by spatial heterogeneity. Sharp edges require high-intensity feature update to

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maintain sharpness; smooth regions need low-intensity feature update; dense details demand progressive refinement to ensure structural consistency (Tai et al., 2017). However, existing methods follow an “input-static mapping-output” paradigm and fail to adapt to this spatial heterogeneity requirement. Traditional CNNs use fixed convolution kernels (Kim et al., 2016) with static receptive fields and weights, unable to adjust feature update rates based on texture type, which limits the restoration of complex textures. Transformers and Mambas capture complex dependencies but rely on static single-time-step feature snapshots (Guo et al., 2025, 2024; Liang et al., 2021), failing to model the continuous texture feature update from initial states to fine-grained restoration and thus causing edge blurring or artifacts. Even lightweight methods for spatial heterogeneity adaptation such as dynamic convolution (Chen et al., 2020; Tian et al., 2024; Wei et al., 2025) and adaptive attention (Hu et al., 2018; Kim et al., 2024; Lin et al., 2024) employ discrete operations and cannot leverage the redundancy and complementarity of features generated by continuous-time feature update, which hinders the exploration of the spatial heterogeneity laws underlying texture restoration.

Liquid neural networks (LNNs) (Hasani et al., 2021) model continuous-time neuron state evolution via ordinary differential equations (ODEs), providing a fundamental technical component support to address the aforementioned bottleneck. They offer three core advantages: a continuous evolution mechanism that aligns with the spatial heterogeneity of texture restoration, where a learnable τ adaptively adjusts feature update rates specific to different textures (fast for edges, slow for smooth regions) to overcome the limitations of static modeling; multi-time-step feature snapshots generated during LNN evolution that enable effective information screening for multi-stage feature fusion; and ODE-based modeling of complex nonlinear relationships and continuous state transitions, which delivers strong expressive capability without the need for deep stacking or complex structures thus supporting enhanced performance under lightweight constraints.

Existing LNN research primarily focuses on sequential tasks such as time series prediction (Hasani et al., 2021), communication systems (Wang et al., 2024), and autonomous driving (Hasani et al., 2022). Its application in computer vision remains limited, and to the best of our knowledge, no prior work has explored LNN-inspired mechanisms for image super-resolution. The core mismatch is that canonical LNNs are designed to model temporal dependencies in sequential data, whereas super-resolution requires modeling spatial texture correlations. Directly transplanting the original LNN formulation cannot deliver effective gains for texture restoration. Even the rare LNN-based visual applications (e.g., image classification Pawlak et al., 2024) are limited to small datasets like CIFAR-10. Ambiguous technical details also hinder reproducibility, which complicates extending LNNs to high-demand tasks like SR.

To address the above challenges and gaps, we migrate the core design insight of the continuous-time evolution of LNNs into image SR and propose LiquidSR, a lightweight framework that offers a new perspective for this task. It adopts a three-stage design: spatial heterogeneity feature extraction via continuous evolution, accurate feature fusion, and spatial heterogeneity long-range dependency modeling, enabling synergistic breakthroughs in both performance and efficiency. First, to leverage the continuous feature update capability of LNNs, we design a Spatial-aware Spatial Heterogeneity Neuron (SSN) module. Based on ordinary differential equations (ODEs), it enables adaptive feature updating from initial states to fine-grained details, adapting to distinct texture restoration needs through a learnable time constant τ . The temporal dimension does not correspond to physical time, but instead represents the evolution trajectory of feature representations. As illustrated in Fig. 1, the LiquidSR with ODE-driven spatial heterogeneity features achieves superior performance compared with those based on CNN, Transformer, or Mamba architectures. Second, to tackle multi-time-step feature redundancy in LNNs, we propose an Adaptive Temporal Attention Weights (ATAW) mechanism. It learns the importance of features across fea-

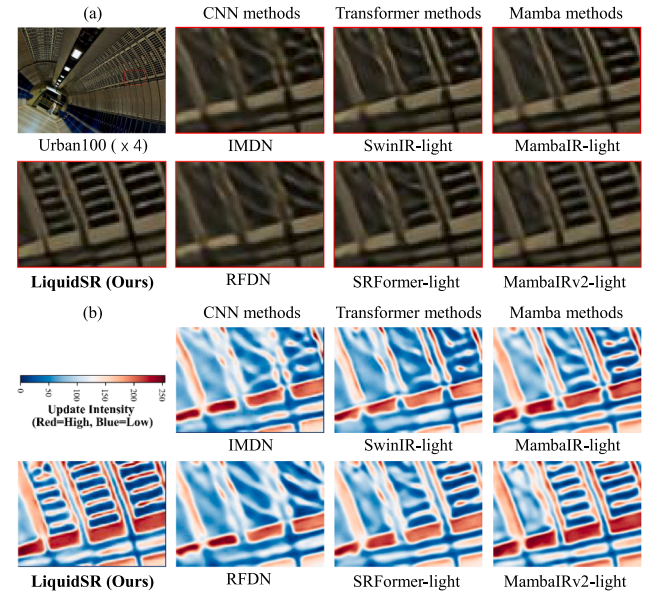


Fig. 1. (a) Local magnification comparison of the reconstruction effects among different modeling schemes. We comprehensively compare the reconstruction performance of three representative static modeling paradigms (CNN, Transformer, and Mamba) with our proposed LiquidSR. Benefiting from ODE-based adaptive feature updating, LiquidSR achieves consistent structure recovery and artifact-free reconstruction. (b) Corresponding update intensity heatmap of (a). Redder color indicates higher update intensity, while bluer color indicates lower update intensity. The heatmap intuitively demonstrates the spatially adaptive distribution of update intensity across different image regions, which effectively validates the strong adaptive update capability of the proposed method.

ture update stages, enhances effective texture information at key time points, suppresses redundancy, and improves fusion efficiency. Finally, to resolve the disconnection between long-range dependency modeling and spatial heterogeneity features in existing lightweight methods, we design a Differentiable Spatial Heterogeneity Self-Attention (DSSA) module. It optimizes feature spatial heterogeneity via ODEs before computing attention on evolved features, synergizing spatial heterogeneity feature update with long-range dependency capture while ensuring the global structural consistency of HR images. As illustrated in Fig. 2, LiquidSR outperforms state-of-the-art efficient SR models with lower complexity and superior performance.

The core reason for choosing ODE-driven continuous feature evolution as the technical route is that it provides a novel paradigm to break through the inherent performance-efficiency trade-off in lightweight image super-resolution. Traditional lightweight SR methods built on the static discrete mapping paradigm face an unavoidable contradiction: improving reconstruction performance requires deeper or wider network architectures, which inevitably increases computational overhead and parameter scale; while pursuing lightweight design will limit feature representation capacity and lead to performance degradation. The continuous dynamical system perspective offers a new solution to this dilemma. It approximates complex nonlinear feature mapping through continuous state transition without deep stacking of network layers, and realizes adaptive adjustment of feature update intensity for different texture regions only through channel-wise learnable time constants with negligible parameter overhead. Compared with traditional CNN, Transformer, Mamba and discrete adaptive schemes, this continuous adaptive paradigm can achieve a favorable balance between reconstruction quality and model complexity, which is of great significance for breaking the performance bottleneck of lightweight SR.

The main contributions of this paper are as follows:

1. We tailor the continuous-time state evolution mechanism of LNNs for spatial texture modeling and propose the SSN module. It models feature refinement via ODEs with learnable channel-wise time constants τ , enabling texture-adaptive update rates to resolve the static modeling bottleneck in lightweight SR.

2. To handle multi-time-step features generated from LNN continuous feature update, we design the ATAW mechanism. It adaptively learns importance weights for different time points, enhances features at key stages, suppresses redundancy, and boosts the effectiveness of feature representation while maintaining the lightweight nature of the model.

3. We propose a DSSA module that enhances feature spatial heterogeneity via the differential equations of spatial heterogeneity neurons prior to modeling long-range dependencies. It retains the long-range capability of attention, improves modeling accuracy through continuous feature update, and addresses the neglect of spatial heterogeneity texture changes in static attention.

2. Related work

2.1. Lightweight image super-resolution

Deep learning-driven lightweight CNN-based image super-resolution (SR) methods (Ahn et al., 2018; Dong et al., 2014; Hui et al., 2019, 2018; Li et al., 2020; Liu et al., 2020; Luo et al., 2020) have achieved notable success. SRCNN (Dong et al., 2014) pioneered the use of a three-layer CNN for end-to-end LR-to-HR mapping, laying the foundation for related research. Subsequent works focus on balancing performance and efficiency, with existing methods constrained by static feature mapping falling into three categories. Convolution kernel optimization methods (FSRCNN Dong et al., 2016, ESPCN Shi et al., 2016) reduce computation via dimension compression and other techniques but lack the ability to adapt to spatial heterogeneity textures. Network structure simplification methods (CARN Ahn et al., 2018, IDN Hui et al., 2018, IMDN Hui et al., 2019) improve parameter efficiency through residual sharing and more but suffer from limited spatial heterogeneity adaptability due to weakened feature interaction. Lightweight variants of emerging architectures (SwinIR-light Liang et al., 2021, ELAN Zhang et al., 2022, PromptSR Liu et al., 2026, MambaIRv2-light Guo et al., 2025) enhance long-range dependency modeling via Transformer/SSM (Gu & Dao, 2023; Vaswani et al., 2017) modifications but rely on static feature snapshots, leading to inefficiency and inadequate adaptability. Recent works further enrich lightweight SR designs from complementary perspectives. DM-Net (Li et al., 2026) explores frequency-domain feature modulation for efficient texture recovery; IHMambaSR (Deng et al., 2026) optimizes state space scanning for 2D image features; PII-SR (Shen et al., 2026) adopts region-differentiated feature integration to improve parameter efficiency. All these methods are rooted in the static feature mapping paradigm, which imposes a ceiling on reconstruction performance under tight parameter budgets.

2.2. Adaptive mechanisms

Modeling spatial texture heterogeneity is a core challenge for image super-resolution. Existing solutions can be broadly categorized into two technical lines: differential equation-driven continuous modeling paradigms and discrete adaptive mechanisms. Neural ordinary differential equations (Neural ODEs) constitute the mainstream continuous modeling path, which can be further divided into ODE-inspired discrete structure design and genuine continuous-depth modeling. The genuine continuous-depth paradigm, which parameterizes network propagation as a continuous differential equation solved via numerical solvers, was formally established by the seminal work of (Chen et al., 2018). For super-resolution tasks, existing continuous-depth works generally adopt

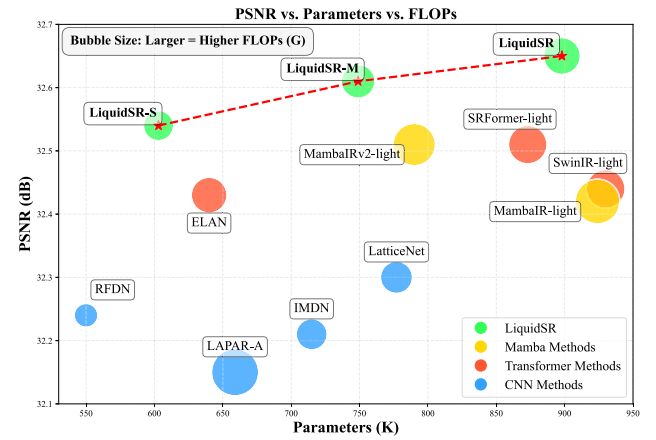


Fig. 2. Performance and model complexity comparison on Set5 dataset for up-scaling factor $\times 4$.

globally shared dynamic parameters and have not introduced channel-wise adaptive evolution rates to accommodate spatial texture heterogeneity (Park & Kim, 2022). Notably, works with controllable restoration strength such as CResMD (He et al., 2020) and ClassSR (Kong et al., 2021) are essentially discrete convolutional architectures; their “continuity” refers to smooth output adjustment rather than continuous dynamical propagation, and thus do not belong to this paradigm. Discrete adaptive mechanisms are more widely adopted, including deformable convolution, dynamic filters and mixture of experts (MoE) (Chen et al., 2020; Li et al., 2022; Yang et al., 2019). However, these schemes face inherent bottlenecks in lightweight scenarios: they all introduce non-negligible parameter and computational overhead via extra prediction branches or parallel expert networks, and their single-step discrete transformation paradigm cannot capture the gradual coarse-to-fine texture refinement process. SR-specific adaptive architectures such as SPIN (Zhang et al., 2023) and Omni-SR (Wang et al., 2023) further extend this idea to full architectural design via superpixel-guided attention partitioning and multi-dimensional feature aggregation respectively, achieving favorable performance-efficiency trade-offs while remaining within the discrete layer-wise propagation framework.

Built on the continuous dynamical system framework, the proposed ODE-based feature update method addresses the above limitations. Benefiting from channel-wise learnable time constants, it achieves adaptive feature update with negligible parameter overhead, models feature refinement as a continuous evolution trajectory to capture gradual texture variation, and can be flexibly embedded into convolution and attention operators as a general state evolution framework.

2.3. Liquid neural networks

Liquid neural networks (LNNs) (Hasani et al., 2021) leverage ordinary differential equations (ODEs) to enable continuous neuron state feature update, delivering spatial heterogeneity adaptability, parameter efficiency, and the ability to capture complex spatial heterogeneity processes. They excel in time-series prediction (Hasani et al., 2021), communication scenarios (Wang et al., 2024), and autonomous driving (Hasani et al., 2022), but research on their application in computer vision remains nascent. Existing work is limited to simple image classification (Pawlak et al., 2024) on small datasets like CIFAR-10, with ambiguous implementations and no application in low-level vision tasks such as image SR. The key gap is that existing LNN feature update functions are designed for time-series dependencies and cannot adapt to the spatial texture modeling requirements of SR. In this work, we explore the integration of liquid neural networks into lightweight image super-resolution and propose LiquidSR, a novel framework that

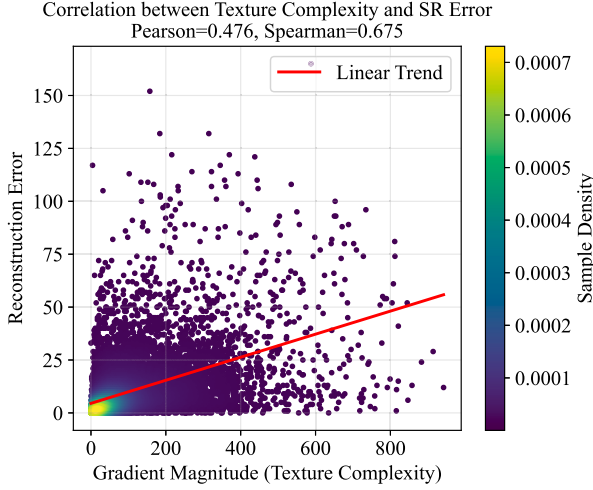


Fig. 3. Relationship between texture complexity and super-resolution reconstruction error. Each point represents a sampled pixel. The horizontal axis denotes the gradient magnitude of the high-resolution image, which reflects local structural complexity. The red line indicates the linear regression trend. The density of points is visualized by color intensity.

leverages ODE-driven continuous feature evolution to adaptively model spatial texture heterogeneity.

It should be clarified that this work does not directly transplant the complete LNN architecture designed for time-series tasks to image super-resolution. Instead, the core connection lies in the migration of the core design insight of LNNs: we inherit its core mechanism of modulating state update rates via learnable time constants, utilizing ODE with learnable per-channel time constants to achieve continuous adaptive state update, and carry out targeted innovations oriented to low-level vision tasks. Specifically, we transform the temporal dynamics modeling for 1D sequential data into feature representation evolution dynamics modeling for 2D spatial textures, and design domain-specific supporting modules to adapt to the spatial heterogeneity modeling requirement of image SR. This work extends the application scope of LNN mechanisms from time series analysis to low-level computer vision, and provides a novel technical paradigm for lightweight image super-resolution.

3. Texture complexity and reconstruction error analysis

Despite significant progress in image super-resolution (SR), accurately reconstructing complex textures and high-frequency structures remains a challenging problem. While modern SR models can produce visually pleasing results in smooth regions, they often struggle to faithfully recover fine structural details in texture-rich areas. Intuitively, regions with higher structural complexity contain more high-frequency information, making them inherently more difficult to reconstruct. However, the relationship between texture complexity and reconstruction error is rarely analyzed explicitly in existing SR studies. Understanding this relationship is important for revealing the limitations of SR models and for guiding the design of methods that better preserve structural details.

Correlation Analysis. To investigate how texture complexity affects SR reconstruction performance, we analyze the relationship between local structural complexity and reconstruction error. Specifically, texture complexity is quantified using the gradient magnitude computed from the high-resolution (HR) image. The reconstruction error is defined as the absolute difference between the SR output and the HR ground truth. As illustrated in Fig. 3, each point represents a sampled pixel from the test image. The horizontal axis denotes the gradient magnitude, which reflects local structural complexity, while the vertical axis indicates the reconstruction error. The results reveal a clear positive correlation be-

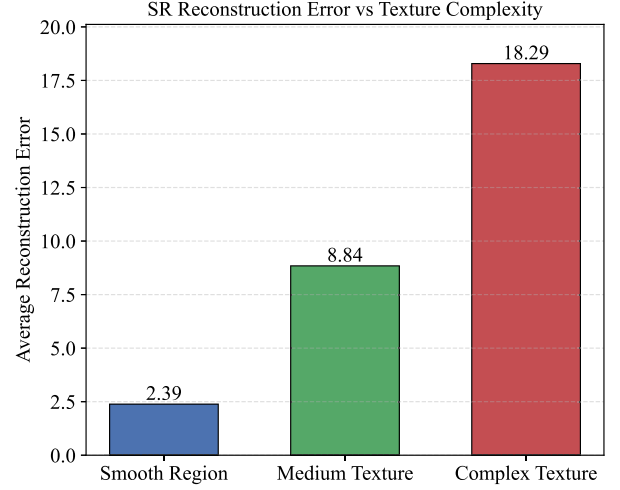


Fig. 4. Average reconstruction error across different texture complexity regions. Pixels are grouped into smooth, medium-texture, and complex-texture regions according to the percentile distribution of gradient magnitude. The results show that reconstruction error increases significantly as texture complexity increases.

tween texture complexity and reconstruction error, with a Pearson correlation coefficient of 0.476 and a Spearman correlation coefficient of 0.675.

Regional Error Statistics. To further quantify this observation, we divide pixels into three groups according to the percentile distribution of gradient magnitude, corresponding to smooth, medium-texture, and complex-texture regions. The average reconstruction error is then computed for each category. As illustrated in Fig. 4, the reconstruction error increases significantly with texture complexity. Smooth regions exhibit the lowest average error (2.39), while medium-texture regions show a higher error (8.84). In highly complex texture regions, the error further increases to 18.29.

The above observations suggest that regions with higher texture complexity require stronger and more adaptive feature updates during reconstruction. However, conventional SR models rely on fixed transformations that cannot dynamically adjust their feature evolution processes. This motivates us to introduce a continuous feature evolution mechanism based on liquid neural networks, where the feature update rate can be adaptively modulated via learnable time constants.

4. Method

4.1. Formal problem statement

Formally, given a low-resolution input $I_{LR} \in \mathbb{R}^{3 \times H \times W}$ and its corresponding high-resolution ground truth $I_{HR} \in \mathbb{R}^{3 \times sH \times sW}$ where s denotes the upscaling factor, lightweight super-resolution aims to learn a mapping function $f(\cdot)$ that minimizes reconstruction error, subject to strict constraints on parameter count Θ and computational cost C :

$$\min_f \mathcal{L}(I_{SR}, I_{HR}) \quad \text{s.t.} \quad |\Theta| \leq \Theta_{\max}, \quad C(f) \leq C_{\max} \quad (1)$$

where $\mathcal{L}(\cdot)$ denotes the pixel-wise reconstruction loss, and $\Theta_{\max}, C_{\max}$ are the pre-defined upper bounds of model complexity.

The intrinsic difficulty of this optimization problem lies in the spatially heterogeneous property of natural textures: the optimal feature update intensity varies across spatial positions. Conventional static paradigms adopt globally uniform mapping parameters, which cannot match this spatially differentiated recovery demand and thus induce an inherent performance-efficiency trade-off. To break this limitation, we model the feature refinement process as an ODE-governed adaptive

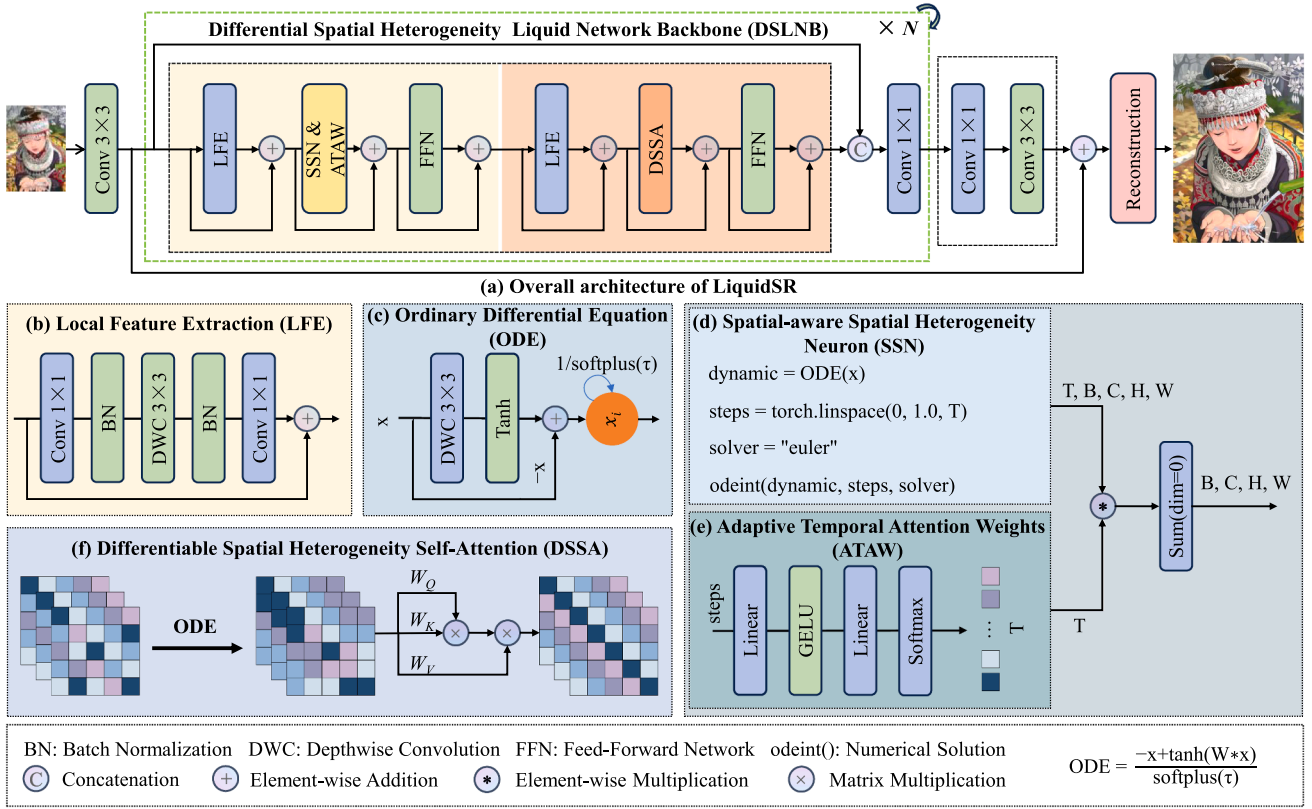


Fig. 5. Illustration of the proposed LiquidSR. (a) Overall architecture of LiquidSR. (b) Architecture of the Local Feature Extraction (LFE). (c) Architecture of the ordinary differential equation (ODE). (d) Architecture of the Spatial-Aware Spatial Heterogeneity Neuron (SSN). (e) Architecture of the Adaptive Temporal Attention Weights (ATAW). (f) Architecture of the Differentiable Spatial Heterogeneity Self-Attention (DSSA).

evolution trajectory, realizing spatially differentiated feature update with negligible extra parameter overhead.

4.2. Overall framework

LiquidSR is a lightweight image super-resolution (SR) framework built upon the ODE-driven continuous feature update mechanism derived from liquid neural networks (LNNs), centered on the adaptive modeling of texture spatial heterogeneity. As illustrated in Fig. 5(a), the framework is designed around the adaptive modeling of texture spatial heterogeneity and consists of four modules: Shallow Feature Extraction (SFE), Differential Spatial Heterogeneity Liquid Network Backbone (DSLNB), feature fusion, and reconstruction. Under the constraints of low parameters and low computational complexity, it achieves accurate reconstruction from Low-Resolution (LR) images to High-Resolution (HR) images. The overall workflow of LiquidSR is as follows: After LR input images are processed by the SFE module, they are fed into the DSLNB backbone to complete three core operations: continuous spatial heterogeneity feature evolution, multi-time-step feature selection and fusion, and heterogeneous long-range dependency modeling. Subsequently, the feature fusion module performs spatial fusion and information enhancement on the deep features output by DSLNB. Finally, the reconstruction module completes upsampling of the specified scale through lightweight PixelShuffle, outputting the SR reconstructed image. Given an LR input image $I_{LR} \in \mathbb{R}^{B \times 3 \times H \times W}$ (where B denotes the batch size, 3 is the number of image channels, and H and W represent the height and width of the LR image, respectively), the SFE module extracts shallow features via a 3×3 convolution:

$$X_{sfe} = W_{3 \times 3} * I_{LR} \quad (2)$$

where $W_{3 \times 3}$ is the 3×3 convolution kernel, $*$ denotes the 2D spatial convolution operation, and $X_{sfe} \in \mathbb{R}^{B \times C \times H \times W}$ (where C is the number of

channels, set to 55 through ablation experiments to balance feature expression and lightweight constraints) is the output shallow feature. This completes the channel dimension transformation from the image pixel space to the feature space. DSLNB is the core feature learning module of LiquidSR and the key carrier integrating LNNs' ODE-driven continuous feature update and spatial heterogeneity modeling. Taking the shallow feature X_{sfe} as input, this module outputs deep features X_{dslnb} through multi-stage feature optimization:

$$X_{dslnb} = \mathcal{F}_{DSLNB}(X_{sfe}) \quad (3)$$

where $\mathcal{F}_{DSLNB}(\cdot)$ represents the feature mapping operation of the DSLNB module. Its detailed internal structure and computational process will be elaborated in Section 4.3. To fully integrate the deep feature information output by DSLNB and enhance the robustness and expressive ability of features, the feature fusion module achieves spatial interaction and fusion of features via a 3×3 convolution. The fused features integrate shallow spatial information and deep heterogeneous features, providing more comprehensive texture support for the reconstruction module:

$$X_{fusion} = W'_{3 \times 3} * X_{dslnb} \quad (4)$$

where $W'_{3 \times 3}$ is the 3×3 convolution kernel (distinguished from the convolution kernel of the SFE module to avoid symbol confusion), and the output fused feature is $X_{fusion} \in \mathbb{R}^{B \times C \times H \times W}$. The reconstruction module is the final output module of LiquidSR, which maps the fused feature tensor from the LR feature space to the SR image space, completing $\times 2 / \times 3 / \times 4$ super-resolution reconstruction. This module adopts PixelShuffle to avoid the surge in computational complexity caused by direct upsampling. Meanwhile, it retains the spatial structure information of initial shallow features through residual connection, alleviating texture blurring in deep reconstruction:

$$I_{SR} = \mathcal{F}_{REC}(X_{fusion} + X_{sfe}) \quad (5)$$

where $\mathcal{F}_{\text{REC}}(\cdot)$ denotes the feature mapping operation of the reconstruction module. It first maps the fused features to the number of channels matching the super-resolution scale via a 3×3 convolution, then completes spatial dimension upsampling through PixelShuffle. Finally, it outputs the SR image $I_{\text{SR}} \in \mathbb{R}^{B \times 3 \times sH \times sW}$, completing the entire super-resolution reconstruction workflow. To minimize the pixel-wise error between the SR reconstructed image and the HR ground-truth image, this paper uses the L1 loss function to train LiquidSR. Given an LR input image I_{LR} and its corresponding HR image I_{HR} , the SR image I_{SR} output by LiquidSR should be similar to I_{HR} . The loss function is defined as:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \|I_{\text{SR}}^{(i)} - I_{\text{HR}}^{(i)}\|_1 \quad (6)$$

where θ represents the parameter set of the entire network, N is the number of LR-HR image pairs used for training, and $I_{\text{SR}}^{(i)}$ and $I_{\text{HR}}^{(i)}$ denote the SR image and HR image of the i th pair, respectively.

4.3. Differential spatial heterogeneity liquid network backbone

As illustrated in Fig. 5(a), the overall workflow of DSLNB is as follows: The shallow feature X_{sfe} is first processed by the Local Feature Extraction (LFE) module to enhance the spatial correlation of local textures, providing a more effective local feature foundation for subsequent ODE evolution. A residual connection is introduced to avoid information loss during local feature interaction and improve training stability:

$$X_{\text{lfe}} = \mathcal{F}_{\text{LFE}}(X_{\text{sfe}}) + X_{\text{sfe}} \quad (7)$$

where $\mathcal{F}_{\text{LFE}}(\cdot)$ is the feature mapping operation of the LFE module. As illustrated in Fig. 5(b), this module is composed of a 1×1 pointwise convolution, a 3×3 depthwise separable convolution, and another 1×1 pointwise convolution in series. A Batch Normalization (BN) function is introduced before the 1×1 pointwise convolution to enhance feature expression and training stability. The output feature X_{lfe} of the LFE module is fed into the Spatial-aware Spatial Heterogeneity Neuron (SSN) module to complete ODE-driven continuous spatial heterogeneity feature evolution, generating multi-time-step features X_{ssn} adapted to different texture regions:

$$X_{\text{ssn}} = \mathcal{F}_{\text{SSN}}(X_{\text{lfe}}) \quad (8)$$

where $\mathcal{F}_{\text{SSN}}(\cdot)$ is the feature evolution operation of the SSN module. Its core is to assign differentiated update rates to different texture regions through a learnable time constant τ , and the specific principle will be elaborated in Section 4.4.

Due to the information redundancy of multi-time-step features X_{ssn} and the varying contributions of features at different time steps to SR reconstruction, the Adaptive Temporal Attention Weights (ATAW) module is introduced to learn the attention weights X_{ataw} for each time step and perform weighted fusion of multi-time-step features. Meanwhile, a residual connection is adopted to fuse the feature X_{lfe} , avoiding the loss of effective information:

$$\begin{aligned} X_{\text{ataw}} &= \mathcal{F}_{\text{ATAW}}(T) \\ X_{\text{sum}} &= \text{Sum}(X_{\text{ssn}} \odot X_{\text{ataw}}) + X_{\text{lfe}} \end{aligned} \quad (9)$$

where $\mathcal{F}_{\text{ATAW}}(\cdot)$ is the attention weight learning operation of the ATAW module, and T is the total number of discrete time steps of the SSN module, whose detailed design will be expanded in Section 4.5. $\odot$ denotes element-wise multiplication of tensors. To achieve dimension matching between weights and feature tensors, X_{ataw} is broadcast to the same dimension as X_{ssn} . $\text{Sum}(\cdot)$ sums the weighted multi-time-step features along the time dimension to obtain the aggregated feature X_{sum} , completing redundant information suppression and effective information enhancement. The aggregated feature X_{sum} passes through the Feed-Forward Network (FFN) module, which realizes refined optimization of feature dimensions through non-linear transformation via a lightweight

design of upscaling, non-linear activation, and downscaling, enhancing the expressive ability of heterogeneous textures:

$$X_{\text{ffn}} = \mathcal{F}_{\text{FFN}}(X_{\text{sum}}) + X_{\text{sum}} \quad (10)$$

where $\mathcal{F}_{\text{FFN}}(\cdot)$ is the FFN module, composed of a 1×1 convolution, GELU activation, and another 1×1 convolution in series. To further strengthen local feature interaction and long-range dependency capture, X_{ffn} undergoes multi-stage feature optimization through the LFE module, Differentiable Spatial Heterogeneity Self-Attention (DSSA) module, and FFN module in sequence. A residual connection is introduced at each stage to ensure the effectiveness of feature transmission:

$$\begin{aligned} X'_{\text{lfe}} &= \mathcal{F}_{\text{LFE}}(X_{\text{ffn}}) + X_{\text{ffn}} \\ X_{\text{dssa}} &= \mathcal{F}_{\text{DSSA}}(X'_{\text{lfe}}) + X'_{\text{lfe}} \\ X'_{\text{ffn}} &= \mathcal{F}_{\text{FFN}}(X_{\text{dssa}}) + X_{\text{dssa}} \end{aligned} \quad (11)$$

where $\mathcal{F}_{\text{DSSA}}(\cdot)$ is the heterogeneous long-range dependency modeling operation of the DSSA module. It integrates ODE-driven spatial heterogeneity feature enhancement with self-attention, improving the adaptability of long-range dependency modeling to texture heterogeneity. The specific principle will be elaborated in Section 4.6. Finally, to fully integrate the initial input and multi-stage optimized features of DSLNB, the shallow feature X_{sfe} output by the SFE module and the final optimized feature X'_{ffn} are concatenated along the channel dimension. A 1×1 convolution is used to complete channel dimension reduction and feature fusion, obtaining the final output feature X_{dslnb} of DSLNB:

$$X_{\text{dslnb}} = W_{1 \times 1}(\text{Concat}(X_{\text{sfe}}, X'_{\text{ffn}})) \quad (12)$$

where $\text{Concat}(\cdot)$ denotes the concatenation operation of tensors along the channel dimension, and $W_{1 \times 1}$ is the 1×1 convolution kernel, used to realize channel dimension unification and feature fusion of the concatenated features. Verified by ablation experiments, this paper stacks 6 DSLNB modules in LiquidSR to achieve the optimal balance between feature expression ability and model computational complexity.

4.4. Spatial-aware spatial heterogeneity neuron

The SSN module serves as the core carrier of continuous feature evolution. It models the feature refinement process via ODEs to realize differentiated update rates across texture types, with a lightweight design that imposes negligible parameter overhead.

As illustrated in Fig. 5(d), as the core mechanism for continuous spatial heterogeneity feature update, the state feature update of the SSN module is mathematically described by ODEs. To explicitly characterize how learnable channel-wise time constants τ adapt restoration dynamics to different textures, we formulate the feature evolution as the following ODE (Fig. 5(c)):

$$\frac{dx}{dt} = \frac{-x + \sigma(W * x)}{\tau} \quad (13)$$

where $x \in \mathbb{R}^{B \times C \times H \times W}$ denotes the input feature tensor (B : batch size, C : number of channels, H, W : height and width of the feature map). W is a depthwise separable convolution kernel that strengthens the spatial correlation between pixels, providing a spatial texture basis for continuous feature update. $\sigma(\cdot)$ is the Tanh activation function, which introduces nonlinearity into the feature update process to enhance feature expressiveness. τ is a learnable time constant and the core of the ODE-based continuous-time feature update, adjusting the feature update rates of distinct textures: a smaller τ accelerates feature update, while a larger τ slows them, enabling texture-specific feature update. $\frac{dx}{dt}$ denotes the speed of feature update; $\sigma(W * x)$ ($*$: convolution) is the feature update target (the ideal feature state after spatial correlation enhancement via W); $-x + \sigma(W * x)$ is the feature update driving force (difference between the ideal state $\sigma(W * x)$ and the current state x). The design of the channel-wise learnable time constant τ is directly inspired by the adaptive time constant mechanism of liquid neural networks. By assigning differentiated evolution rates to different feature channels, we realize

adaptive feature update matching the spatial heterogeneity of textures, which constitutes a key extension of the standard Neural ODE paradigm in this work.

Rationale for continuous dynamical modeling. It is worth clarifying that the temporal dimension t in the proposed ODE formulation does not correspond to physical time in time-series tasks. Instead, it parameterizes the evolution trajectory of feature representations on a high-dimensional feature manifold. This design is justified from both empirical and theoretical perspectives:

Empirical evidence. The correlation analysis in Section 3 demonstrates that reconstruction error rises monotonically with texture complexity. This implies that the ideal feature refinement process is spatially non-uniform: complex texture regions (e.g., edges, dense patterns) require progressive multi-step updates to recover fine details, while smooth regions only need mild transformation. Such non-uniform refinement is inherently a continuous dynamic process where feature states evolve from coarse to fine along the depth of representation, rather than a one-step discrete mapping. **Theoretical rationality.** From the perspective of dynamical systems, the successive feature states within the SSN module can be viewed as discrete sampling points along an underlying continuous evolution trajectory. ODE parameterizes this trajectory into a continuous differential form, where the learnable time constant τ controls the effective step size of each channel on the feature manifold. Specifically, smaller τ yields larger update steps for fast refinement of complex textures, while larger τ yields smaller steps for stable preservation of smooth regions.

Obtaining the analytical solution to Eq. (13) is intractable in practice. We thus employ numerical discretization to convert the continuous feature evolution into iteratively computable steps. Common ODE solvers include the Euler method, Runge-Kutta method (RK4), and the Midpoint method. To select the solver best suited to our lightweight design goal, we conduct ablation studies to compare both reconstruction performance and computational overhead across different schemes. The results show that the Euler method meets lightweight requirements, achieving competitive accuracy with maximum computational efficiency. It is thus selected as the numerical solution for the SSN module, converting the continuous feature update into an iterative computation with T discrete time steps, as follows:

$$x(t_i) = x(t_{i-1}) + \Delta t \cdot \frac{-x(t_{i-1}) + \sigma(W * x(t_{i-1}))}{\tau} \quad (14)$$

where $t_i = i \times \Delta t$ ($i = 1, 2, \dots, T$), with Δt (discrete time-step size, set to 0.5) and $T = 2$. This configuration ensures sufficient texture restoration during continuous feature update while avoiding redundant computation and artifacts. The feature update process starts from initial features, undergoes intermediate spatial heterogeneity updates at $t_1 = 0.5$, and generates detail-rich evolved features at $t_2 = 1.0$, simulating the texture restoration process from the initial state to fine-grained details. To stabilize the feature update process, the convolution kernel W uses orthogonal initialization to prevent gradient explosion. Empirically, we observe that $T=2$ achieves the best performance, providing sufficient feature evolution while avoiding over-smoothing and excessive computational overhead, as shown in Fig. 6 and Table 12.

Theoretical properties of the liquid ODE. We first analyze the mathematical properties of the continuous dynamical system defined in Eq. (13). The proposed ODE formulation belongs to the class of Lipschitz-continuous dynamical systems. Since the depthwise convolution operator W and the tanh activation function are both Lipschitz-continuous, the right-hand side of Eq. (13) satisfies the Lipschitz condition, which guarantees the existence and uniqueness of the solution for any initial feature state $x(0)$.

Building on its Lipschitz continuity, the system is globally asymptotically stable with a unique smooth fixed point under reasonable parameter initialization, and any initial feature state converges toward this

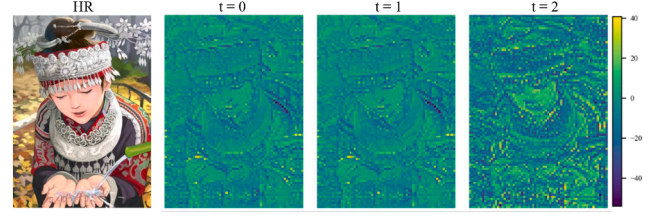


Fig. 6. Feature evolution in the proposed SSN. The visualization shows the feature responses at different evolution steps. As the evolution progresses from $t=0$ to $t=2$, structural patterns such as edges and textures become progressively more pronounced.

steady state as evolution time increases. For the adopted forward Euler discretization, numerical stability is jointly determined by the unit time step and the learnable time constant τ . The end-to-end learnable τ automatically remains within a numerically stable interval during training, ensuring reliable convergence of the evolution process.

Starting from the Euler discretization form given in Eq. (14), we can rearrange the update rule into the following equivalent residual form:

$$x(t_i) = \left(1 - \frac{\Delta t}{\tau}\right)x(t_{i-1}) + \frac{\Delta t}{\tau}\sigma(W * x(t_{i-1})) \quad (15)$$

This formulation can be interpreted as a generalized residual block with an adaptive residual weight $\alpha = \Delta t/\tau$. Unlike standard residual networks with fixed residual weights, α here is modulated by the channel-wise learnable τ , enabling channel-adaptive feature update intensity. This adaptive mechanism is theoretically more capable of fitting the non-uniform recovery requirement of spatially heterogeneous textures than fixed residual structures.

Design considerations for lightweight implementation. The adoption of the Euler solver and $T = 2$ discrete steps is a task-specific design tailored for lightweight super-resolution, rather than an incomplete simplification of liquid neural networks. The rationality of this configuration is explained from three perspectives:

Efficiency constraint of lightweight scenarios. High-order solvers (e.g., RK4) and more evolution steps will cause a linear increase in computational overhead, which runs counter to the core goal of lightweight SR. As verified in subsequent ablation studies, increasing T beyond 2 does not yield consistent performance gains; instead, it introduces over-smoothing effects and redundant computation. The Euler + $T = 2$ configuration achieves the optimal balance between feature representation capability and computational efficiency.

Numerical suitability for shallow non-stiff systems. The proposed Liquid ODE is a non-stiff system with shallow evolution depth. High-order solvers such as RK4 are originally designed to suppress severe error accumulation in stiff systems and long-trajectory evolution. For our shallow non-stiff system, the redundant high-order correction terms in RK4 will introduce unnecessary numerical perturbations, which disrupt smooth feature state transition and degrade reconstruction performance. In contrast, the Euler method with only one function evaluation per step has better numerical stability for small-step shallow evolution.

Cost-effectiveness of channel-independent τ . Pixel-level or spatial-position-level τ would introduce massive parameter overhead and increase the risk of overfitting, which is not suitable for lightweight tasks. Channel-independent τ corresponds to different texture feature patterns, and realizes differentiated update rates for various texture types with only C learnable parameters ($C = 55$ in our default setting). Ablation results confirm that this design brings significant performance improvement with negligible parameter increase.

4.5. Adaptive temporal attention weights

The SSN module leverages the ODE-based continuous-time feature update of LNNs to output multi-time-step features

$X = [x(t_1), x(t_2), \dots, x(t_T)]$. As illustrated in Fig. 1, these features contribute differently to SR reconstruction: initial-time-step features lack spatial heterogeneity information and have limited value for detail restoration; intermediate features (e.g., $x(t_1)$) undergo partial feature update, retain transient texture details, and serve as a key basis for capturing texture variation trends; later-stage features (e.g., $x(t_2)$) stabilize with integrated evolved details, and are critical for complete texture restoration. Existing strategies have limitations: using only the last-time-step feature ignores intermediate transient details and results in the loss of spatial heterogeneity information; simple concatenation of all features fails to highlight key information and introduces redundant interference. To address these issues, this paper designs an ATAW mechanism that accurately selects and fuses key intermediate and later-stage features while dynamically suppressing redundancy, thereby improving the efficiency and accuracy of feature fusion for SR reconstruction.

As illustrated in Fig. 5(e), the ATAW mechanism aims to accurately screen multi-time-step evolved features, and consists of two calculation steps. First, the temporal attention weights are learned: the nonlinear features of time points are captured via a lightweight adaptive temporal attention mechanism to generate attention weights, as follows:

$$w = \text{Softmax}(\text{FC}_2(\text{GELU}(\text{FC}_1(t)))) \quad (16)$$

where $t = [t_1, t_2, \dots, t_T]$ is the time-step vector, corresponding to different stages of ODE-based continuous feature update. FC_1 and FC_2 are linear layers (hidden dimension = 16) that map each t_i to a 16-dimensional vector to capture nonlinear features at each time point. GELU is the activation function, and Softmax ensures a normalized, interpretable weight distribution. Second, weighted fusion of multi-time-step features is performed: the learned attention weights are applied to these features to generate fused features, as follows:

$$x_{\text{weighted}} = \sum_{i=1}^T w_i \cdot x(t_i) \quad (17)$$

where $w = [w_1, w_2, \dots, w_T]$ denotes temporal attention weights ($\sum_{i=1}^T w_i = 1$). A larger w_i indicates a greater contribution of the i th time-step feature to SR reconstruction. The fusion process automatically focuses on effective features at key feature update stages, suppresses redundant information, and enhances the targeting and effectiveness of feature expression. $x(t_i)$ is the evolved feature at the i th time-step. The ATAW module adopts a lightweight design with only two linear layers, which barely increases the model's computational burden. It automatically adjusts attention weights based on the texture characteristics of different images, ensuring a reasonable weight distribution via Softmax normalization. This avoids the excessive dominance of features at a single time point and improves the model's generalization ability, whose effectiveness is verified by ablation studies.

4.6. Differentiable spatial heterogeneity self-attention

Long-range dependency modeling is essential for preserving global structural consistency. To align attention computation with texture-adaptive feature representations, we design the DSSA module (Fig. 5(f)) that integrates ODE-based feature enhancement into self-attention computation. Its design idea is to first enhance feature spatial heterogeneity via ODEs, then compute self-attention on evolved spatial heterogeneity features to make long-range dependency modeling more flexible and adaptive to texture variations. The DSSA workflow is as follows: first, weighted fused features output by ATAW x_{weighted} are fed into ODEs (consistent with Eq. (13) in the SSN module) to enhance spatial heterogeneity expression. Unlike the SSN module, this feature update dispenses with numerical solutions (e.g., Euler's method) and directly generates features for subsequent modeling, capturing subtle spatial heterogeneity changes in textures and outputting spatial heterogeneity features $x_{\text{dyn}} \in \mathbb{R}^{B \times C \times H \times W}$. Then, self-attention is computed based on x_{dyn} ,

with the specific Eq. as follows:

$$\text{Attn}(x_{\text{dyn}}) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (18)$$

where $Q, K, V = x_{\text{dyn}} \cdot W_Q, x_{\text{dyn}} \cdot W_K, x_{\text{dyn}} \cdot W_V$ (with W_Q, W_K, W_V as projection matrices) and d_k denotes the query dimension. The module first enhances feature spatial heterogeneity via ODE feature update, then captures long-range dependencies through self-attention to improve the accuracy of attention calculation. Evolved spatial heterogeneity features contain richer texture details, enabling attention weights to accurately model long-range dependencies and ensure the global structural consistency of SR images.

5. Experiments

5.1. Datasets and evaluation metrics

In this study, the widely adopted DIV2K dataset (Timofte et al., 2017), which consists of 800 pairs of high-resolution (HR) and low-resolution (LR) images, was used to train the model on an NVIDIA GeForce RTX 3080 GPU. To comprehensively evaluate the performance of the proposed method, systematic experiments were conducted on five standard test sets: Set5 (Bevilacqua et al., 2012), Set14 (Zeyde et al., 2010), BSD100 (Martin et al., 2001), Urban100 (Huang et al., 2015), and Manga109 (Matsui et al., 2017). All experiments strictly followed standard practices in the field. Quantitative evaluations were performed on the luminance (Y) channel of the YCbCr color space, with Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) used as evaluation metrics.

5.2. Implementation details

The training of LiquidSR was conducted on the NVIDIA GeForce RTX 3080 GPU. During the model training phase, a random patch-based sampling strategy was adopted for training batch construction, where each low-resolution (LR) input image was randomly cropped into 16 patches of size 64×64 . This strategy ensures training efficiency while fully preserving local contextual information. To further enhance the generalization ability of the model, geometric data augmentation techniques were incorporated, including random $0^\circ/90^\circ/180^\circ/270^\circ$ rotations and horizontal/vertical flipping, all applied synchronously to paired LR-HR training samples. The LiquidSR network features a lightweight architectural design, with the number of channels set to 55 and the number of DSLNB modules fixed at 6. These specific hyperparameters were determined via cross-validation (see the ablation experiments in Section 5.4 for the impacts of channel count and DSLNB module number). The Adam optimizer was employed for the optimization process, with hyperparameters configured as $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e-8$ to balance training stability and convergence speed. The total number of training epochs was 1000, with an initial learning rate of 5×10^{-4} that was halved every 200 epochs. The aforementioned settings effectively balance the model's training stability and final performance.

5.3. Comparison methods

We compare our LiquidSR with representative super-resolution methods: the bicubic interpolation baseline and advanced lightweight architectures (IDN (Hui et al., 2018), CARN (Ahn et al., 2018), LAPAR-A (Li et al., 2020), IMDN (Hui et al., 2019), RFDN (Liu et al., 2020), LatticeNet (Luo et al., 2020), SwinIR-light (Liang et al., 2021), ELAN (Zhang et al., 2022), SRFormer-light (Zhou et al., 2023), MambaIR-light (Guo et al., 2024), SPIN (Zhang et al., 2023), Omni-SR (Wang et al., 2023), CATANet (Liu et al., 2025), MCSR (Chen et al., 2026), MambaIRv2-light (Guo et al., 2025)). All experiments follow a standardized reproducible protocol: we use the official public implementations of all compared methods, identical training-test splits, and consistent evaluation metrics

Table 1

Quantitative comparison on lightweight image super-resolution with state-of-the-art methods. The best and the second best results are in **bold** and underline, respectively.

Method	Scale	Params (K)	FLOPs (G)	Set5		Set14		BSD100		Urban100		Manga109	
				PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic		–	–	33.66	0.9299	30.24	0.8688	29.56	0.8431	26.88	0.8403	30.80	0.9339
IDN (Hui et al., 2018)		553	124.6	37.83	0.9600	33.30	0.9148	32.08	0.8985	31.27	0.9196	38.01	0.9749
CARN (Ahn et al., 2018)		1592	222.8	37.76	0.9590	33.52	0.9166	32.09	0.8978	31.92	0.9256	38.36	0.9765
LAPAR-A (Li et al., 2020)		548	171.0	38.01	0.9605	33.62	0.9183	32.19	0.8999	32.10	0.9283	38.67	0.9772
IMDN (Hui et al., 2019)		694	158.8	38.00	0.9605	33.63	0.9177	32.19	0.8996	32.17	0.9283	38.88	0.9774
RFDN (Liu et al., 2020)		534	95.0	38.05	0.9606	33.68	0.9184	32.16	0.8994	32.12	0.9278	38.88	0.9773
LatticeNet (Luo et al., 2020)		756	169.5	38.15	0.9610	33.78	0.9193	32.25	0.9005	32.43	0.9302	38.94	0.9773
SwinIR-light (Liang et al., 2021)		910	244.2	38.14	0.9611	33.86	0.9206	32.31	0.9012	32.76	0.9340	39.12	0.9783
ELAN (Zhang et al., 2022)	× 2	621	203.1	38.17	0.9611	33.94	0.9207	32.30	0.9012	32.76	0.9340	39.11	0.9782
MambaIR-light (Guo et al., 2024)		905	334.2	38.13	0.9610	33.95	0.9208	32.31	0.9013	32.85	0.9349	39.20	0.9782
SRFormer-light (Zhou et al., 2023)		853	236.3	38.23	0.9613	33.94	0.9209	32.36	0.9019	32.91	0.9353	39.28	<u>0.9785</u>
SPIN (Zhang et al., 2023)		497	–	38.20	0.9615	33.90	0.9215	32.31	0.9015	32.79	0.9340	39.18	0.9784
Omni-SR (Wang et al., 2023)		772	–	38.22	0.9613	33.98	0.9210	32.36	0.9020	33.05	0.9363	39.28	0.9784
CATANet (Liu et al., 2025)		477	–	<u>38.28</u>	<u>0.9617</u>	33.99	0.9217	<u>32.37</u>	<u>0.9023</u>	<u>33.09</u>	<u>0.9372</u>	<u>39.37</u>	0.9784
MCSR (Chen et al., 2026)		806	173.7	38.19	0.9612	33.88	0.9206	32.32	0.9016	32.79	0.9342	–	–
MambaRv2-light (Guo et al., 2025)		774	286.3	38.26	0.9615	<u>34.09</u>	<u>0.9221</u>	32.36	0.9019	<u>33.26</u>	<u>0.9378</u>	39.35	<u>0.9785</u>
LiquidSR (Ours)		881	216.9	38.32	<u>0.9616</u>	34.13	<u>0.9224</u>	32.41	0.9025	33.26	0.9383	39.46	0.9786
Bicubic		–	–	30.39	0.8682	27.55	0.7742	27.21	0.7385	24.46	0.7349	26.95	0.8556
IDN (Hui et al., 2018)		553	56.3	34.11	0.9253	29.99	0.8354	28.95	0.8013	27.42	0.8359	32.71	0.9381
CARN (Ahn et al., 2018)		1592	118.8	34.29	0.9255	30.29	0.8407	29.06	0.8034	28.06	0.8493	33.50	0.9440
LAPAR-A (Li et al., 2020)		544	114.0	34.36	0.9267	30.34	0.8421	29.11	0.8054	28.15	0.8523	33.51	0.9441
IMDN (Hui et al., 2019)		703	71.5	34.36	0.9270	30.32	0.8417	29.09	0.8046	28.17	0.8519	33.61	0.9445
RFDN (Liu et al., 2020)		541	42.2	34.41	0.9273	30.34	0.8420	29.09	0.8050	28.21	0.8525	33.67	0.9449
LatticeNet (Luo et al., 2020)		765	76.3	34.53	0.9281	30.39	0.8424	29.15	0.8059	28.33	0.8538	33.63	0.9441
SwinIR-light (Liang et al., 2021)	× 3	918	111.2	34.62	0.9289	30.54	0.8463	29.20	0.8082	28.66	0.8624	33.98	0.9478
ELAN (Zhang et al., 2022)		629	90.1	34.61	0.9288	30.55	0.8463	29.21	0.8081	28.69	0.8624	34.00	0.9478
MambaIR-light (Guo et al., 2024)		913	148.5	34.63	0.9288	30.54	0.8459	29.23	0.8084	28.70	0.8631	34.12	0.9479
SRFormer-light (Zhou et al., 2023)		861	105.4	34.67	0.9296	30.57	0.8469	29.26	0.8099	28.81	0.8655	34.19	0.9489
SPIN (Zhang et al., 2023)		569	–	34.65	0.9293	30.57	0.8464	29.23	0.8089	28.71	0.8627	34.24	0.9489
Omni-SR (Wang et al., 2023)		780	–	34.70	0.9294	30.57	0.8469	<u>29.28</u>	0.8094	28.84	0.8656	34.22	0.9487
CATANet (Liu et al., 2025)		550	–	<u>34.75</u>	<u>0.9300</u>	30.67	0.8481	<u>29.28</u>	<u>0.8101</u>	<u>29.04</u>	<u>0.8689</u>	<u>34.40</u>	<u>0.9499</u>
MCSR (Chen et al., 2026)		823	78.8	34.67	0.9294	30.58	0.8467	29.25	0.8095	28.78	0.8644	–	–
MambaRv2-light (Guo et al., 2025)		781	126.7	34.71	0.9298	<u>30.68</u>	<u>0.8483</u>	29.26	0.8098	29.01	<u>0.8689</u>	<u>34.41</u>	0.9497
LiquidSR (Ours)		888	96.4	34.78	0.9301	30.69	0.8485	29.30	0.8106	29.07	0.8699	34.51	0.9502
Bicubic		–	–	28.42	0.8104	26.00	0.7027	25.96	0.6675	23.14	0.6577	24.89	0.7866
IDN (Hui et al., 2018)		553	32.3	31.82	0.8903	28.25	0.7730	27.41	0.7297	25.41	0.7632	29.41	0.8942
CARN (Ahn et al., 2018)		1592	90.9	32.13	0.8937	28.60	0.7806	27.58	0.7349	26.07	0.7837	30.47	0.9084
LAPAR-A (Li et al., 2020)		659	94.0	32.15	0.8944	28.61	0.7818	27.61	0.7366	26.14	0.7871	30.42	0.9074
IMDN (Hui et al., 2019)		715	40.9	32.21	0.8948	28.58	0.7811	27.56	0.7353	26.04	0.7838	30.45	0.9075
RFDN (Liu et al., 2020)		550	23.9	32.24	0.8952	28.61	0.7819	27.57	0.7360	26.11	0.7858	30.58	0.9089
LatticeNet (Luo et al., 2020)		777	43.6	32.30	0.8962	28.68	0.7830	27.62	0.7367	26.25	0.7873	30.54	0.9073
SwinIR-light (Liang et al., 2021)	× 4	930	63.6	32.44	0.8976	28.77	0.7858	27.69	0.7406	26.47	0.7980	30.92	0.9151
ELAN (Zhang et al., 2022)		640	54.1	32.43	0.8975	28.78	0.7858	27.69	0.7406	26.54	0.7982	30.92	0.9150
MambaIR-light (Guo et al., 2024)		924	84.6	32.42	0.8977	28.74	0.7847	27.68	0.7400	26.52	0.7983	30.94	0.9135
SRFormer-light (Zhou et al., 2023)		873	62.8	32.51	0.8988	28.82	0.7872	27.73	0.7422	26.67	0.8032	31.17	0.9165
SPIN (Zhang et al., 2023)		555	–	32.48	0.8983	28.80	0.7862	27.70	0.7415	26.55	0.7998	30.98	0.9156
Omni-SR (Wang et al., 2023)		792	–	32.49	0.8988	28.78	0.7859	27.71	0.7415	26.64	0.8018	31.02	0.9151
CATANet (Liu et al., 2025)		535	46.8	32.58	0.8998	28.90	0.7880	<u>27.75</u>	<u>0.7427</u>	<u>26.87</u>	<u>0.8081</u>	<u>31.31</u>	<u>0.9183</u>
MCSR (Chen et al., 2026)		846	45.6	32.50	0.8982	28.83	0.7874	27.71	0.7416	26.60	0.8012	–	–
MambaRv2-light (Guo et al., 2025)		790	75.6	32.51	0.8992	28.84	0.7878	<u>27.75</u>	0.7426	26.82	0.8079	31.24	0.9182
LiquidSR (Ours)		898	57.3	32.65	0.9002	28.94	0.7892	27.79	0.7436	26.92	0.8107	31.45	0.9197

Table 2

Quantitative results of three independent training runs (× 4).

Methods	Params (K)	FLOPs (G)	Set5		BSD100		Urban100	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Run 1	898	57.3	32.62	0.9001	27.78	0.7433	26.92	0.8106
Run 2	898	57.3	32.66	0.9001	27.77	0.7434	26.91	0.8104
Run 3 (reported)	898	57.3	32.65	0.9002	27.79	0.7436	26.92	0.8107

Table 3

Results of paired two-tailed t-tests against MambaRv2-light.

Dataset	Number of images	t statistic	Degrees of freedom	p-value	Significance level
Urban100	100	3.13	99	0.0023	$p < .01$
Manga109	109	7.53	108	< 0.001	$p < .001$

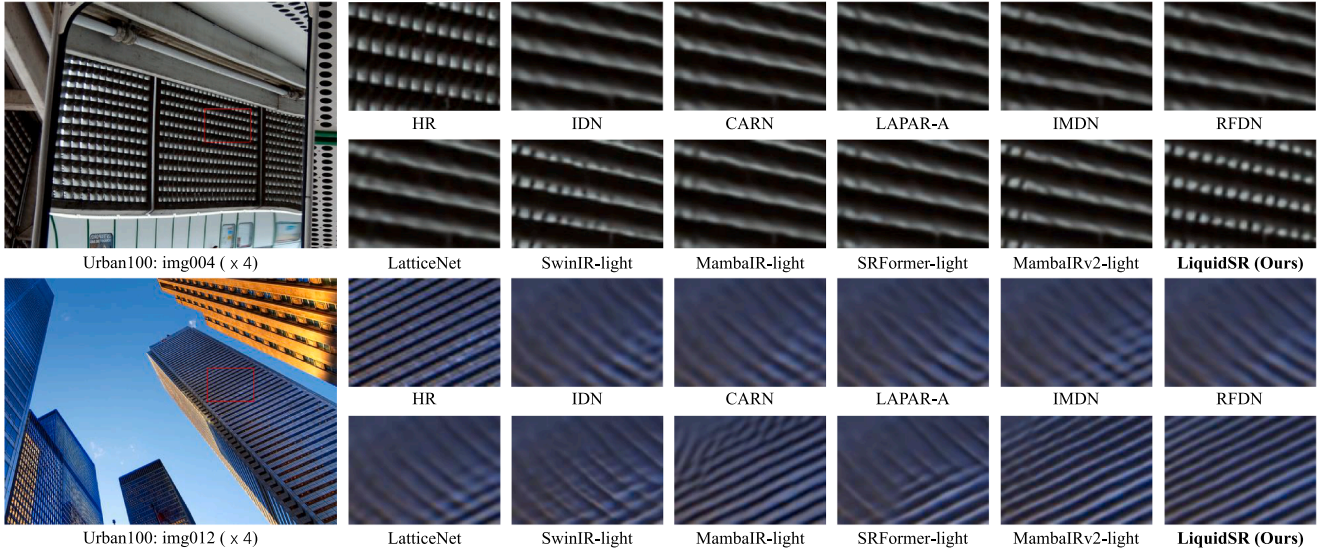


Fig. 7. Qualitative comparison of our LiquidSR with different methods on Urban100 $\times 4$ lightweight image SR.

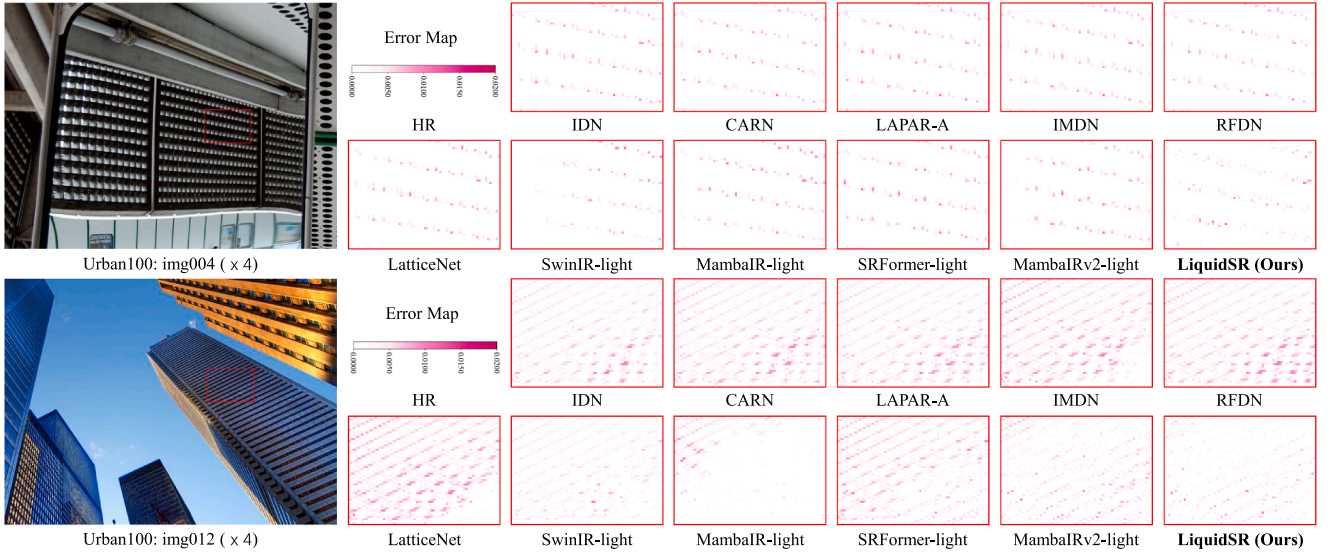


Fig. 8. Comparison of error maps for $\times 4$ lightweight image SR on Urban100. The error maps are computed as the absolute difference between the reconstructed image and the ground truth image. Brighter regions indicate larger reconstruction errors.

to ensure fair and reliable comparisons. All baseline methods have been validated in prior work and represent the state of the art. The effectiveness of LiquidSR is analyzed in detail through quantitative results, visual quality, and model efficiency.

Quantitative Comparison: Table 1 presents objective quantitative results of different methods across five benchmark datasets. Mamba-based architectures outperform traditional CNNs, leveraging the linear long-range dependency modeling advantage of Mamba. Our LiquidSR achieves optimal performance across nearly all datasets and upscaling factors ($\times 2$, $\times 3$, $\times 4$). On Manga109, characterized by dense textures and high demands for detail restoration, LiquidSR outperforms the second-ranked MambaIRv2-light by 0.12 dB, 0.10 dB, and 0.21 dB in $\times 2$, $\times 3$, and $\times 4$ tasks, respectively. This validates the effectiveness of its core modules, particularly the ODE continuous-time feature update of the SSN in boosting reconstruction accuracy. Table 1 also shows LiquidSR breaks performance barriers under lightweight constraints: it has fewer parameters than SwinIR-light and lower computational complexity than MambaIRv2-light,

thanks to ODE's parameter efficiency (approximating complex transformations via continuous feature update without deep stacking) and the lightweight design of ATAW/DSSA. The two variants of LiquidSR in Table 4 also achieve strong performance while maintaining lightweight characteristics. Overall, experimental results confirm LiquidSR achieves state-of-the-art performance in lightweight super-resolution.

To verify the stability and reproducibility of the experimental results, we conducted three independent training runs with different random seeds (Table 2). The standard deviation of PSNR across runs is 0.021 dB on Set5, 0.010 dB on BSD100, and 0.006 dB on Urban100, all of which are far smaller than the observed performance gains over state-of-the-art baselines. This confirms that the training process of the proposed method is highly stable, and the performance improvements are not caused by random fluctuation across runs.

Paired two-tailed t-tests on per-image metric values further confirm the statistical significance of the performance improvement. On Urban100 and Manga109 (Table 3), the advantages of the proposed

Table 4

Quantitative comparison with state-of-the-art lightweight methods for $\times 4$ image super-resolution. The best and the second best results are in **red** and **blue**, respectively.

Method	Scale	Params (K)	FLOPs (G)	Set5		Set14		BSD100		Urban100		Manga109	
				PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic	–	–	–	28.42	0.8104	26.00	0.7027	25.96	0.6675	23.14	0.6577	24.89	0.7866
IDN (Hui et al., 2018)	553	32.3	31.82	0.8903	28.25	0.7730	27.41	0.7297	25.41	0.7632	29.41	0.8942	
CARN (Ahn et al., 2018)	1592	90.9	32.13	0.8937	28.60	0.7806	27.58	0.7349	26.07	0.7837	30.47	0.9084	
LAPAR-A (Li et al., 2020)	659	94.0	32.15	0.8944	28.61	0.7818	27.61	0.7366	26.14	0.7871	30.42	0.9074	
IMDN (Hui et al., 2019)	715	40.9	32.21	0.8948	28.58	0.7811	27.56	0.7353	26.04	0.7838	30.45	0.9075	
RFDN (Liu et al., 2020)	550	23.9	32.24	0.8952	28.61	0.7819	27.57	0.7360	26.11	0.7858	30.58	0.9089	
LatticeNet (Luo et al., 2020)	777	43.6	32.30	0.8962	28.68	0.7830	27.62	0.7367	26.25	0.7873	30.54	0.9073	
SwinIR-light (Liang et al., 2021)	930	63.6	32.44	0.8976	28.77	0.7858	27.69	0.7406	26.47	0.7980	30.92	0.9151	
ELAN (Zhang et al., 2022)	640	54.1	32.43	0.8975	28.78	0.7858	27.69	0.7406	26.54	0.7982	30.92	0.9150	
MambaIR-light (Guo et al., 2024)	× 4	924	84.6	32.42	0.8977	28.74	0.7847	27.68	0.7400	26.52	0.7983	30.94	0.9135
SRFormer-light (Zhou et al., 2023)	873	62.8	32.51	0.8988	28.82	0.7872	27.73	0.7422	26.67	0.8032	31.17	0.9165	
SPIN (Zhang et al., 2023)	555	–	32.48	0.8983	28.80	0.7862	27.70	0.7415	26.55	0.7998	30.98	0.9156	
Omni-SR (Wang et al., 2023)	792	–	32.49	0.8988	28.78	0.7859	27.71	0.7415	26.64	0.8018	31.02	0.9151	
CATANet (Liu et al., 2025)	535	46.8	32.58	0.8998	<u>28.90</u>	0.7880	27.75	0.7427	26.87	0.8081	31.31	0.9183	
MambaRv2-light (Guo et al., 2025)	790	75.6	32.51	0.8992	28.84	0.7878	27.75	0.7426	26.82	0.8079	31.24	0.9182	
LiquidSR-S (Ours)	603	38.5	32.54	0.8989	28.87	0.7866	27.74	0.7413	26.72	0.8040	31.21	0.9166	
LiquidSR-M (Ours)	749	47.8	32.61	0.8995	28.86	0.7884	27.77	0.7431	<u>26.88</u>	0.8096	31.39	0.9193	
LiquidSR (Ours)	898	57.3	32.65	0.9002	28.94	0.7892	27.79	0.7436	26.92	0.8107	31.45	0.9197	

Table 5

Perceptual quality comparison on $\times 4$ benchmark datasets. Lower values indicate better performance.

Method	BSD100		Urban100		Manga109	
	LPIPS	DISTS	LPIPS	DISTS	LPIPS	DISTS
SwinIR-light	0.3675	0.2033	0.2202	0.1714	0.1261	0.0845
MambaIR-light	0.3671	0.2056	0.2214	0.1715	0.1278	0.0865
CATANet	0.3630	0.2031	0.2128	0.1676	0.1228	0.0846
MambaRv2-light	0.3614	0.2027	0.2072	0.1649	0.1216	0.0817
LiquidSR (Ours)	0.3595	0.2007	0.2048	0.1616	0.1215	0.0807

method over the baseline reach significant levels ($t = 3.13$, $df = 99$, $p = 0.0023$ and $t = 7.53$, $df = 108$, $p < 0.001$, respectively), verifying the systematic superiority of the proposed framework.

Perceptual Quality Evaluation. Beyond pixel-wise reconstruction metrics (PSNR and SSIM), we further conduct perceptual quality evaluation to validate the actual visual fidelity of the reconstructed images. We adopt two widely recognized full-reference perceptual metrics, namely Learned Perceptual Image Patch Similarity (LPIPS) and Deep Image Structure and Texture Similarity (DISTS), where lower values indicate better perceptual quality and more natural visual effects. Evaluations are performed on three standard benchmark datasets (BSD100, Urban100, Manga109) under the $\times 4$ super-resolution setting, and the results are summarized in Table 5.

Quantitative results demonstrate that our LiquidSR consistently achieves the best perceptual performance across all three datasets and both metrics, outperforming all compared state-of-the-art lightweight SR methods. Specifically, compared with the previous strongest baseline MambaRv2-light, LiquidSR reduces LPIPS and DISTS by 0.5% and 1.0% on BSD100, 1.2% and 2.0% on Urban100, and 0.1% and 1.2% on Manga109, respectively. Notably, more significant gains appear on texture-rich datasets (Urban100 and Manga109), which corroborates that the continuous evolution mechanism better captures fine-grained texture patterns than static paradigms. The consistent improvement on both pixel-wise metrics and perceptual metrics jointly verifies that the performance gain is not limited to numerical pixel error optimization, but also delivers tangible improvement in visual perception quality.

Qualitative Comparison: We conducted a qualitative comparison between LiquidSR and state-of-the-art lightweight super-resolution methods, with visual results illustrated in Fig. 7. LiquidSR consistently reconstructs sharper edges and finer texture details than all compared lightweight methods, with fewer artifacts and blurring effects. On

Table 6

The average inference time on Urban100 dataset.

Model	SwinIR-light	MambaIR-light	MambaRv2-light	LiquidSR
Latency (ms)	210.5	194.3	362.5	105.4

“img004” and “img012” of Urban100 (Fig. 7), the edge contours reconstructed by LiquidSR are more accurate, while those from other models suffer from blurring, distortion, or discontinuity. This advantage stems from the SSN module, which assigns fast feature update rates to edges for sharpness preservation, and the DSSA module, which captures edge long-range dependencies to avoid discontinuity. Visual results confirm the core advantages of LiquidSR: the SSN adapts to spatial heterogeneity textures via ODE-based continuous feature update, the ATAW focuses on key features, and the DSSA strengthens long-range dependency modeling. Their synergy yields reconstructed images with superior overall consistency and detail fidelity.

To better analyze the spatial distribution of reconstruction errors, we visualize the error maps between the super-resolution images and the ground truth images. The error map is computed as the absolute difference between the reconstructed image and the ground truth image. Brighter regions correspond to larger reconstruction errors. For better visualization, the error values are amplified by a constant factor. As illustrated in Fig. 8, LiquidSR exhibits noticeably fewer bright regions compared with other methods, indicating that it produces smaller reconstruction errors, particularly in areas with complex textures and structural details.

To further evaluate the reconstruction quality between our proposed LiquidSR model and other state-of-the-art methods, we present visual comparison examples in Figs. 9 and 10. The results demonstrate that our approach enjoys clear advantages in recovering sharp edges and fine textures from severely degraded low-resolution inputs. Notably, the LiquidSR model excels in capturing edge details, yielding reconstructions that are not only more accurate but also visually more plausible.

Latency Comparison: We further evaluate the inference latency of all compared methods on the Urban100 dataset. All measurements are performed on a single NVIDIA GeForce RTX 3080 GPU, under FP32 precision and batch size = 1 (single-image inference). The majority of input LR images are at 256×256 resolution, corresponding to 1024×1024 output under the $\times 4$ upscaling setting. For each test image, we conduct 10 repeated inference runs after sufficient warm-up, and remove outliers before calculating the average latency. Quantitative results (Table 6) show that LiquidSR achieves an average inference time of 105.4 ms for

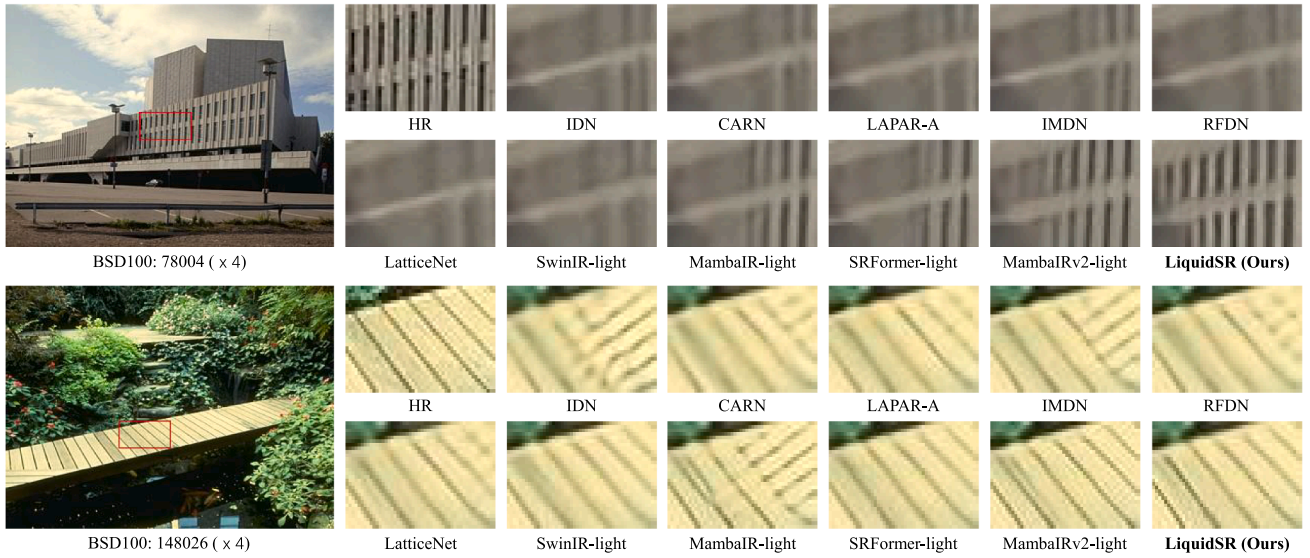


Fig. 9. Qualitative comparison of our LiquidSR with different methods on BSD100 $\times$ 4 lightweight image SR.

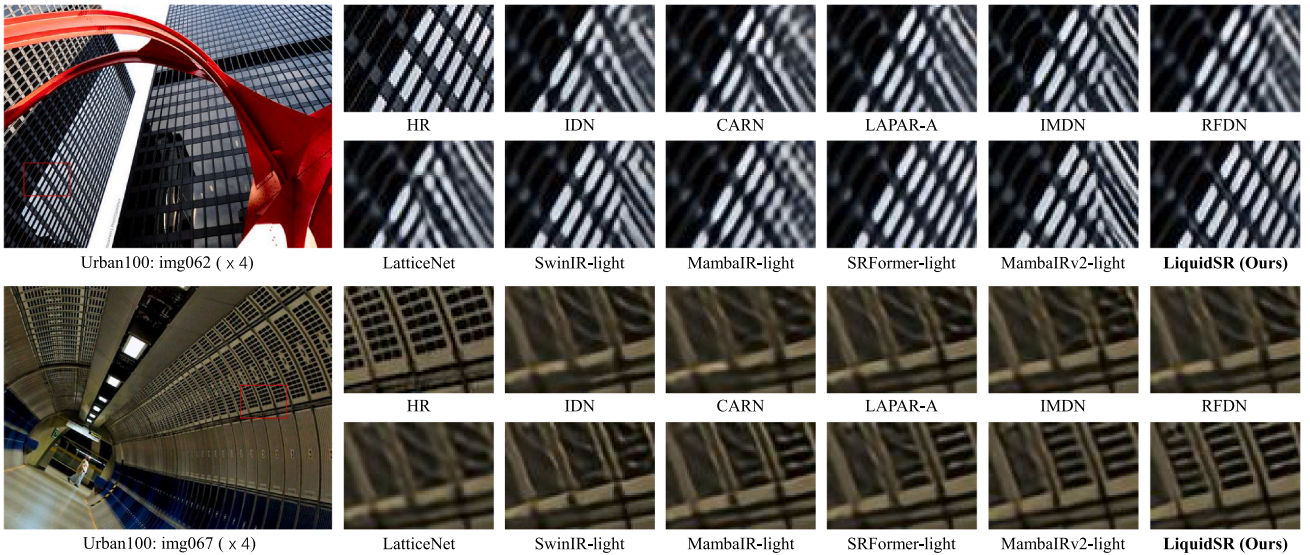


Fig. 10. Qualitative comparison of our LiquidSR with different methods on Urban100 $\times$ 4 lightweight image SR.

Table 7
Peak training GPU memory comparison of different lightweight SR methods.

Model	Peak Training GPU Memory (MB)
MambaIR-light	17,340
MambaIRv2-light	18,152
LiquidSR-S	8362
LiquidSR-M	10078
LiquidSR	11834

1024 $\times$ 1024 output on a single RTX 3080 GPU, which is significantly faster than most compared lightweight SR methods. In terms of training cost, the peak training GPU memory usage of LiquidSR is 11834 MB, approximately 34.8% lower than that of MambaIRv2-light, demonstrating favorable deployment efficiency on consumer-grade GPU platforms (Table 7). We note that all latency measurements are conducted on general-purpose GPU hardware without task-specific deployment opti-

mization; edge-device performance may vary. Targeted on-device optimization and edge deployment will be investigated in our future work.

5.4. Ablation study

Effectiveness of Different Components: As core components of LiquidSR, the SSN, ATAW, and DSSA modules were validated via five ablation settings (Table 8). Using the SSN module alone outperforms convolutional baselines (Table 1), confirming ODE-based feature update enhances feature representation. Integrating ATAW boosts Urban100 performance by 0.06 dB in PSNR, as its adaptive weighted fusion filters key time-step features and suppresses redundancy. Adding DSSA yields PSNR improvements of 0.16 dB (BSD100), 0.61 dB (Urban100), and 0.68 dB (Manga109). This demonstrates that the synergy between spatial heterogeneity feature update and self-attention strengthens long-range dependency modeling, particularly for dense and edge textures, while preserving the lightweight design and global feature consistency. Individual removal of each core module further validates the necessity of all three components.

Table 8
Ablation on the effectiveness of different components.

SSN	ATAW	DSSA	Params (K)	FLOPs (G)	BSD100		Urban100		Manga109	
					PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
✓	✗	✗	743	45.2	27.62	0.7375	26.25	0.7809	30.75	0.9117
✓	✓	✗	744	45.2	27.63	0.7376	26.31	0.7921	30.77	0.9117
✓	✓	✓	898	57.3	27.79	0.7436	26.92	0.8107	31.45	0.9197
✗	✓	✓	743	45.2	27.76	0.7425	26.83	0.8077	31.32	0.9183
✓	✗	✓	897	57.3	27.79	0.7431	26.90	0.8101	31.47	0.9197

Table 9
Ablation on the effectiveness of different solvers.

Solver	Params (K)	FLOPs (G)	Set5		Set14		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
RK4	898	58.4	32.59	0.8994	28.92	0.7880	31.43	0.9190
Midpoint	899	57.7	32.60	0.8996	28.93	0.7885	31.44	0.9194
Euler	898	57.3	32.65	0.9002	28.94	0.7892	31.45	0.9197

Table 10
Ablation on the effectiveness of the ODE feature update.

Methods	Params (K)	FLOPs (G)	Set14		BSD100		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Control-SR	898	57.3	28.90	0.7880	27.77	0.7431	31.33	0.9183
LiquidSR (Ours)	898	57.3	28.94	0.7892	27.79	0.7436	31.45	0.9197

Effectiveness of Different Solvers: Quantitative results are summarized in Table 9. The Euler method achieves the best reconstruction performance across all benchmarks with the lowest computational overhead, while higher-order solvers (Midpoint, RK4) deliver no performance gain and even introduce slight degradation. This phenomenon can be explained from two complementary perspectives. First, for our shallow evolution system with only $T = 2$ steps, the higher numerical precision of RK4 brings negligible marginal benefit, while its multi-stage function evaluations introduce extra intermediate nonlinear perturbations that disrupt the smoothness of feature evolution. Second, with nearly matched parameter counts, the more complex computation path of high-order solvers increases the implicit model capacity, which raises the risk of overfitting under the fixed training budget and leads to degraded generalization performance. This law is consistent with the ablation results of evolution step T , jointly indicating that there exists an optimal complexity threshold for the ODE evolution system. Overall, the Euler method with only one function evaluation per step achieves a favorable balance between numerical efficiency, feature smoothness and generalization capability, which is more compatible with the lightweight SR task.

To isolate the independent contribution of the LNN architecture, we design a control model with matched parameters/FLOPs (denoted as Control-SR) by ablating the ODE feature update module. The Control-SR is designed with exactly the same overall architecture, channel number, module count and all other components (LFE, FFN, DSSA, etc.) as LiquidSR. The only difference is that the ODE-based feature update in the SSN module is replaced with a standard residual block of the same convolution configuration: $x_{out} = x_{in} + \sigma(W * x_{in})$, where W is the same depthwise separable convolution kernel as in the ODE formulation. This setup ensures that the parameter quantity and computational complexity of Control-SR are exactly matched with LiquidSR, so as to isolate the independent contribution of the ODE continuous evolution mechanism. LiquidSR outperforms Control-SR across all benchmarks (Table 10), which validates the unique value of the ODE feature update mechanism for dynamic texture restoration.

Importantly, the multi-step Euler discretization reuses the same set of convolution weights across all iterations, resulting in an identical convolution parameter count to a single residual block. With only negligi-

ble overhead introduced by the learnable per-channel time constants, Control-SR and LiquidSR are strictly matched in total parameters and computational complexity. The observed performance gap arises solely from the ODE formulation: the continuous dynamical system paradigm that frames feature refinement as an evolution trajectory, combined with channel-adaptive update rates modulated by learnable time constants. This controlled setting rigorously isolates the intrinsic value of the ODE design, eliminating the confounding effect of increased parameters or stacked layers.

Effectiveness of Time-Step T : The time-step T determines the discrete steps of ODE-based continuous feature update and directly affects the sufficiency of feature update and computational overhead. We tested five T configurations ($T=1, 2, 3, 4, 5$; Table 12), with the model achieving optimal performance at $T=2$. With $T=1$, the feature evolution is too shallow to fully model heterogeneous texture patterns, resulting in degraded reconstruction accuracy. For $T>2$ ($T=3, 4, 5$), there is no performance improvement or even a decline: excessive time-steps introduce redundant computations, may induce feature overfitting or over-feature update, and generate artifacts. As illustrated in Fig. 11, the characteristic evolution result when $T=5$. Computational complexity increases monotonically with T (56.97 G FLOPs at $T=1$, 58.34 G FLOPs at $T=5$), since each time-step requires ODE feature update computation, and more steps increase overhead.

To further quantitatively verify the over-smoothing effect of excessive time steps and clarify the evolution law of feature details, we calculate the average spatial gradient magnitude of output feature maps at different T values via the standard Sobel operator. Before calculation, all feature maps are first globally min-max normalized to eliminate the interference of value scale drift during continuous ODE iteration, so that the gradient magnitude purely reflects the richness of high-frequency information in features. It should be noted that gradient magnitude includes both valid texture gradients and high-frequency noise artifacts, so higher values do not necessarily correspond to better reconstruction quality. The quantitative results are summarized in Table 11.

The evolution of gradient magnitude presents two distinct stages with different physical meanings:

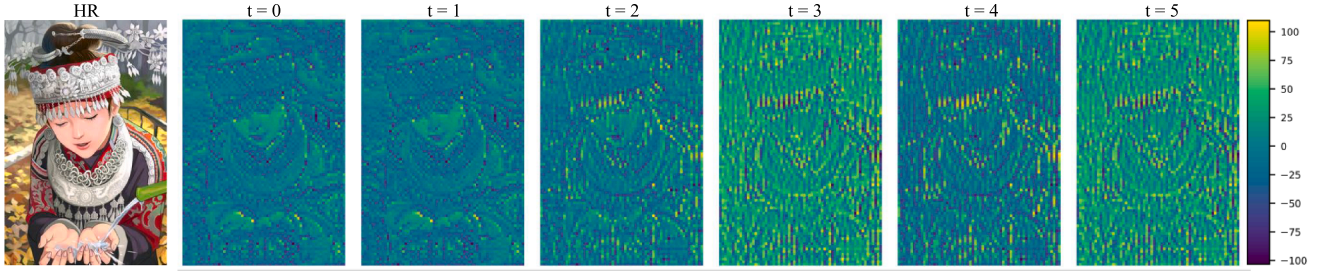


Fig. 11. Visualization of Liquid ODE feature evolution with $T=5$. Despite increasing the evolution steps to $T=5$, the features do not converge to a better state; instead, texture diffusion and redundant activations are observed. This verifies that excessive time steps introduce redundant computation, leading to feature overfitting or over-updating, which ultimately causes performance degradation and artifacts, justifying the rationality of selecting $T=2$ as the optimal discrete step count in this paper.

Table 11

Average spatial gradient magnitude of feature maps under different time steps T .

Number of Time Steps T	1	2	3	4	5
Spatial Gradient Magnitude	0.2966	0.1415	0.1142	0.1179	0.1121

Noise suppression stage ($T=1 \rightarrow T=2$). At $T=1$, features only undergo one extremely shallow update and retain abundant high-frequency noise and artifacts from shallow input features, leading to an artificially high gradient magnitude. From $T=1$ to $T=2$, the ODE completes the first round of sufficient feature evolution, which mainly suppresses invalid noise and regularizes feature distribution. The significant drop in gradient magnitude in this stage is a beneficial feature purification process, which corresponds to a notable improvement in PSNR/SSIM reconstruction performance.

Over-smoothing stage ($T=2 \rightarrow T=5$). After $T=2$, high-frequency noise has been fully suppressed. As T continues to increase from 2 to 5, the average gradient magnitude decreases from 0.1415 to 0.1121, with a relative reduction of 20.8%. This continuous decline stems from the gradual loss of valid high-frequency texture details, i.e., the over-smoothing effect, which leads to the degradation of reconstruction performance.

From the perspective of dynamical system theory, this behavior further validates the mathematical consistency of our formulation instead of contradicting it. The proposed Liquid ODE is a Lipschitz-continuous stable dynamical system with a unique smooth fixed point. As evolution time increases, feature states converge continuously toward this steady state, which causes the overall gradient magnitude to decrease with minor numerical fluctuations from discrete Euler discretization. The optimal reconstruction performance at $T=2$ marks the optimal trade-off between noise suppression and fine detail preservation on the continuous trajectory. This is precisely the core advantage of continuous dynamical modeling: it parameterizes the entire feature evolution path as a differentiable continuous function, allowing us to select the optimal evolution depth flexibly to match task-specific restoration demands.

Combined with the reconstruction performance results, $T=2$ exactly achieves the optimal balance between sufficient noise removal and complete effective detail preservation, thus delivering the best PSNR/SSIM performance. Taking both reconstruction quality and computational cost into consideration, we select $T=2$ as the default configuration.

Effectiveness of the τ Learning Method: The learning method for the time constant τ determines its adaptability to diverse texture restoration requirements. We compared two approaches: channel-shared (the same τ for all channels) and channel-independent (τ learned independently per channel; Table 13). The channel-independent τ achieves 31.45 dB PSNR on Manga109, 0.23 dB higher than the channel-shared τ (31.32 dB). This confirms that distinct channels correspond to unique

Table 12

Ablation on the effectiveness of time-step T .

T	Params (K)	FLOPs (G)	Set5		Set14		BSD100	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
1	898	57.0	32.60	0.8997	28.93	0.7889	27.78	0.7434
2	898	57.3	32.65	0.9002	28.94	0.7892	27.79	0.7436
3	898	57.7	32.59	0.8999	28.93	0.7891	27.78	0.7435
4	898	58.1	32.60	0.8995	28.88	0.7877	27.78	0.7429
5	898	58.3	32.57	0.8999	28.90	0.7884	27.78	0.7434

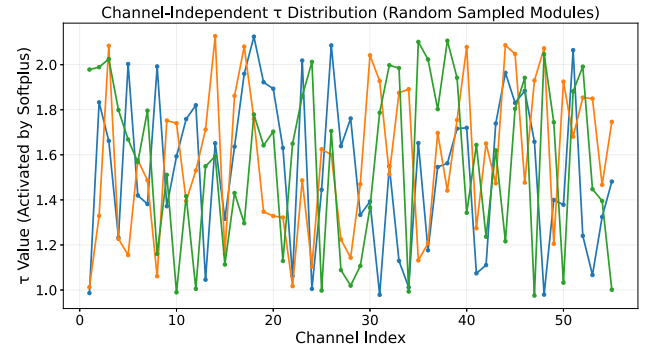


Fig. 12. Three modules are randomly selected from the model to demonstrate the value distribution of the channel-independent time constant τ . Each feature channel corresponds to a unique τ value, whose positivity is guaranteed by the Softplus activation; the differentiated distribution of τ intuitively reflects the texture-adaptive update strategy learned by the model. The abscissa represents the channel index within each module, and the ordinate denotes the τ value after Softplus activation.

texture features, and the channel-independent method learns optimal feature update rates for each channel. The channel-shared τ fails to adapt to such differentiated demands, leading to performance degradation. Fig. 12 validates the rationality and effectiveness of the channel-independent τ design. Thus, we adopt the channel-independent τ learning method to enable precise adaptation to diverse texture features.

Comparison with Discrete Adaptive Mechanisms: To verify the parameter efficiency advantage of the proposed ODE-based adaptive mechanism, we conduct comparative experiments with three representative discrete adaptive schemes under the same backbone architecture: DynamicConv2d (dynamic filter) (Chen et al., 2020), CondConv2d (MoE-based dynamic convolution) (Yang et al., 2019), and ODConv2d (multi-dimensional dynamic convolution) (Li et al., 2022). All methods replace the standard convolution in the baseline with the corresponding adaptive module, and keep the rest of the network structure unchanged.

The results are summarized in Table 14. All compared discrete adaptive methods introduce significant parameter overhead but fail to achieve superior reconstruction performance. Specifically, our method

Table 13
Ablation on the effectiveness of the τ learning method.

τ	Params (K)	FLOPs (G)	BSD100		Urban100		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Channel-shared	898	57.0	27.77	0.7430	26.89	0.8099	31.32	0.9185
Channel-independent	898	57.3	27.79	0.7436	26.92	0.8107	31.45	0.9197

Table 14
Comparison with discrete adaptive mechanisms on $\times 4$ benchmark datasets.

Method	Params (K)	BSD100		Urban100		Manga109	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DynamicConv2d	2211	27.77	0.7430	26.92	0.8094	31.40	0.9188
CondConv2d (MoE)	2204	27.77	0.7425	26.87	0.8090	31.38	0.9181
ODConv2d	1239	27.77	0.7427	26.88	0.8091	31.39	0.9183
LiquidSR (Ours)	898	27.79	0.7436	26.92	0.8107	31.45	0.9197

reduces parameters by 59.4% compared with DynamicConv2d and 27.5% compared with ODConv2d, while achieving the best performance on all three benchmarks. This confirms that the continuous evolution paradigm achieves stronger adaptive modeling capability at a far lower parameter cost, making it more favorable for lightweight SR settings.

Effectiveness of the ATAW Fusion Method: To verify the advantages of the ATAW, we compared its performance and efficiency with average fusion (fuses all time-step features after concatenation; Table 15). ATAW achieves 32.65 dB PSNR on Set5 (0.10 dB higher than the 32.55 dB of average fusion) and 31.45 dB on Manga109 (0.08 dB higher than the 31.37 dB of average fusion). This confirms ATAW adaptively enhances key time-step features and suppresses redundancy, while average fusion dilutes critical information, leading to performance degradation. ATAW has fewer parameters (898 K, approximately 7.5% lower than the 971 K of average fusion) and lower computational complexity (57.3 G FLOPs, approximately 7.3% lower than the 61.8 G FLOPs of average fusion). This is because ATAW learns weights via two lightweight linear layers, whereas average fusion requires extra feature concatenation and processing that increase overhead. In conclusion, ATAW improves performance while reducing overhead, making it a superior feature fusion method.

Effectiveness of DSSA: To clarify the advantages of Differentiable Spatial Heterogeneity Self-Attention (DSSA) over conventional Self-Attention (SA), we designed a control experiment that compares their performance and efficiency under the same network architecture, with only the attention module replaced (Table 16). DSSA outperforms SA across all metrics: it achieves 32.65 dB PSNR on Set5 (0.04 dB higher than the 32.61 dB of SA), and exhibits a more pronounced advantage on dense-texture Manga109 (0.09 dB higher than the 31.36 dB of SA). This performance gain stems from DSSA optimizing feature spatial heterogeneity update via ODEs before computing attention; it captures texture spatial heterogeneity to enrich feature information, unlike SA, which relies on static features. In terms of efficiency, DSSA (898 K parameters, 57.3 G FLOPs) is nearly identical to SA (895 K parameters, 56.8 G FLOPs). Only minimal overhead is introduced by lightweight ODE feature update, demonstrating that DSSA balances performance improvement and efficiency preservation, aligning with the core requirements of lightweight super-resolution.

To further isolate the specific contribution of the ODE mechanism and rule out the impact of simply increasing parameters, we conducted controlled experiments by replacing the ODE branch in DSSA with standard convolution layers. Experimental results (Table 17) show that under comparable or even lower parameter and computational budgets, the ODE-based feature enhancement scheme achieves significantly better performance than standard convolution

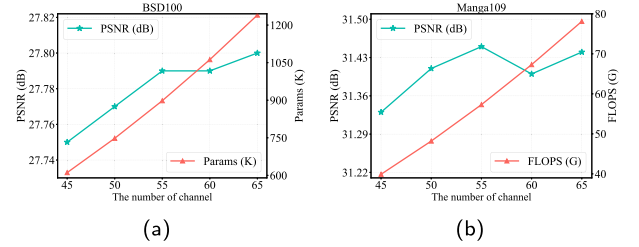


Fig. 13. Ablation on the effectiveness of channel number.

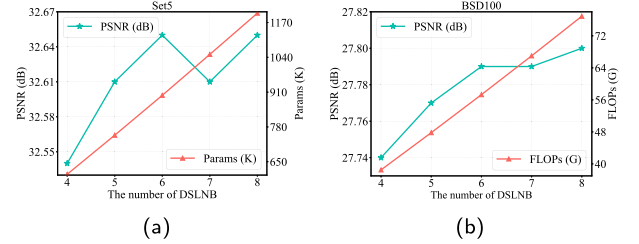


Fig. 14. Ablation on the effectiveness of DSLNB number.

layers, which verifies that the continuous feature evolution mechanism has higher structural efficiency and is the core source of performance improvement.

Effectiveness of the Number of Channels: The number of channels determines the feature representation capability and parameter scale of a model. We tested five configurations (45, 50, 55, 60, 65; Fig. 13). Performance increases monotonically as the number of channels rises from 45 to 55: The PSNR of Manga109 increases from 31.22 dB to 31.45 dB, and that of BSD100 from 27.74 dB to 27.79 dB. When channels exceed 55 (60, 65), performance saturates or degrades. This is because excessive channels introduce redundant features, increase the risk of overfitting, and fail to further enhance representation capability. The parameter count and computational complexity increase monotonically with the number of channels. To match the complexity of mainstream lightweight methods (e.g., SwinIR-light Liang et al., 2021, MambaIRv2-light Guo et al., 2025) in complexity while balancing performance and efficiency, the default number of channels is set to 55. It is a common practice in lightweight SR research to use non-power-of-two channel counts for precise complexity control, as also adopted in representative works such as SwinIR-light (60 channels) and MambaIRv2-light (48 channels). The candidate channel range from 45 to 65 is selected to align with the parameter scale of mainstream lightweight SR baselines (approximately 500K to 1M parameters), ensuring the search is conducted within a fair comparison interval. The optimal value of 55 channels reaches the best trade-off between feature representation capacity and model complexity under our 6-module architecture: fewer channels fail to capture sufficient texture details, while more channels introduce redundant parameters and increase overfitting risk. It is worth noting that all ablation experiments are performed on the validation split of the DIV2K training set, and the 55-channel configuration consistently achieves superior performance on multiple unseen test benchmarks, further validated by three independent training runs with different random seeds, which show highly stable performance with PSNR standard

Table 15
Ablation on the effectiveness of ATAW fusion method.

Fusion Method	Params (K)	FLOPs (G)	Set5		Set14		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Average Fusion	971	61.8	32.55	0.8993	28.84	0.7874	31.37	0.9187
ATAW	898	57.3	32.65	0.9002	28.94	0.7892	31.45	0.9197

Table 16
Ablation on the effectiveness of the DSSA.

Module	Params (K)	FLOPs (G)	Set5		Set14		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
SA	892	57.0	32.61	0.8997	28.89	0.7884	31.36	0.9189
DSSA	898	57.3	32.65	0.9002	28.94	0.7892	31.45	0.9197

Table 17
Ablation study on the specific contribution of the ODE mechanism in the DSSA module ($\times 4$ super-resolution setting). We replace the ODE feature enhancement branch with standard convolution layers of different configurations to isolate the effect of continuous evolution from parameter and computation increment.

Module	Params (K)	FLOPs (G)	Set5		Set14		Manga109	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Replace ODE with 1×1 conv	929	59.2	32.61	0.8997	28.83	0.7849	31.40	0.9192
Replace ODE with 3×3 conv	1219	77.1	32.67	0.9003	28.91	0.7888	31.44	0.9194
3×3 conv with matched param budget	845	53.8	32.50	0.8989	28.86	0.7871	31.24	0.9178
LiquidSR (Ours)	898	57.3	32.65	0.9002	28.94	0.7892	31.45	0.9197

deviations below 0.021 dB across all benchmarks, which verifies its generalization and rules out over-tuning to the validation set.

Effectiveness of the Number of DSLNB Modules: The number of DSLNB modules determines a model's depth and feature processing capability. We tested five configurations (4, 5, 6, 7, 8; Fig. 14). Performance increases monotonically as the number of modules rises from 4 to 6: The PSNR of Set5 increases from 32.54 dB to 32.65 dB, and that of BSD100 from 27.74 dB to 27.79 dB. When the number of modules exceeds 6 (7, 8), performance saturates: the PSNR of Set5 drops to 32.63 dB, and that of BSD100 remains at 27.79 dB. This is because excessive modules cause feature over-processing, introduce redundancy, and increase the risk of gradient vanishing. The parameter count and computational complexity increase monotonically with the number of modules. To balance performance and efficiency while keeping the scale comparable to mainstream lightweight methods, the default number of DSLNB modules is set to 6.

6. Limitations

Despite the favorable performance-efficiency trade-off of LiquidSR for lightweight image super-resolution, several limitations remain and warrant further investigation. To satisfy lightweight constraints, the proposed method adopts Euler discretization with only two evolution steps for the ODE system, which does not fully exploit the theoretical representation capacity of continuous dynamical systems; further increasing the number of evolution steps will lead to accumulated numerical errors and gradual over-smoothing of features. Furthermore, all efficiency benchmarks are tested on desktop GPUs, and actual on-device performance including inference latency, memory footprint and energy consumption on edge hardware remains to be verified with dedicated hardware-specific optimization. Finally, the current adaptive mechanism operates at the channel level, which cannot achieve finer pixel-level spatial adaptation for heterogeneous textures within the same feature channel.

7. Conclusion

To address the limitations of static modeling in texture-adaptive lightweight super-resolution, this paper presents LiquidSR, a framework that integrates continuous ODE-driven feature evolution inspired by liquid neural networks. Its key components are the SSN module, ATAW mechanism, and DSSA module. The SSN models feature update via ODEs with a learnable τ , the ATAW learns adaptive weights for key time-steps to suppress redundancy, and the DSSA synergizes ODE-based feature update with long-range dependency modeling. Experimental results show LiquidSR achieves state-of-the-art performance across five public datasets while maintaining low parameters, low computational complexity, and fast inference speed. These validate the effectiveness of ODE-based continuous feature update for lightweight SR. Future work will extend this study in three complementary aspects aligned with the aforementioned limitations. We will first develop a texture-adaptive time-step mechanism for the ODE system to mitigate over-smoothing and numerical error accumulation, fully unleashing the representation potential of continuous dynamical systems under lightweight constraints. Second, we will extend the current channel-level adaptive mechanism to finer spatial granularity for more precise pixel-level heterogeneous texture modeling. Finally, we will conduct deployment optimization on typical edge devices (e.g., ARM-based embedded platforms) to verify practical inference latency, memory footprint and energy consumption in resource-constrained scenarios.

CRedit authorship contribution statement

Hanjiang Lin: Conceptualization, Methodology, Software, Writing – original draft; **Huiying Xu:** Supervision, Writing – review & editing; **Hongbo Li:** Supervision, Writing – review & editing; **Wenjie Wang:** Validation, Formal analysis, Visualization; **Wenzhe Tan:** Validation, Formal analysis, Visualization; **Tao Guan:** Validation, Formal analysis, Visualization; **Chao Chen:** Validation, Formal analysis, Visualization; **Zhendong Chen:** Validation, Formal analysis, Visualization; **Yux-**

iang Zhang: Validation, Formal analysis, Visualization; **Yanxue Liang:** Validation, Formal analysis, Visualization; **Xinzhong Zhu:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of generative AI and AI-assisted technologies

Generative AI tools were used solely for language polishing in the writing process. All academic content and conclusions are original work of the authors, who take full responsibility for the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgments

This work was supported by the [National Natural Science Foundation of China](#) (62376252); Zhejiang Province Province-Land Synergy Program (2025SDXT004-3); Zhejiang Province Leading Geese Plan (2025C02025, 2025C01056).

Data availability

Data will be made available on request.

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